From Wallets to Wages: Consumer Income, Job Design, and Pay Disparities

Soohyun Roh *

July 20, 2026

[Click here for the latest version]

Abstract

Pay differences between organizations are a key source of wage inequality. I offer a novel perspective on these differences by theorizing that firms targeting consumers at different income levels systematically create different kinds of jobs. Firms targeting high-income consumers specialize work into higher-paying, higher-skilled positions oriented toward quality, whereas firms building their operations around lower-income consumers emphasize cost minimization by assigning workers a broader range of general tasks. Matching consumer foot traffic data and establishment-level wage records, I find that establishments serving higher-income consumers pay their workers more. I then exploit the pandemic-era collapse in commuting as a shock to establishments' customer bases and find that losing affluent consumers led exposed establishments to lower pay, even after extensive controls for changes in customer volume. Analysis of online job postings further reveals that jobs at higher-income-serving firms involve a narrower set of tasks that command higher market value. These findings show how consumer markets shape firms' internal job design and contribute to pay inequality across organizations in the service economy.

*MIT Sloan. Email: rohs@mit.edu.

1 Introduction

What drives pay inequality between firms? Research has documented substantial variation in pay across employers and attributed much of the growth in wage inequality over the past four decades to widening pay differences between firms (Barth et al. 2016; Song et al. 2019). Organizational research has long examined how firms exercise meaningful discretion over pay, partly independent of market forces (Avent-Holt and Tomaskovic-Devey 2014; Ton 2014). Yet the generalizability of these accounts remains uncertain (Osterman 2018), particularly as external market forces have assumed a greater role in wage setting (Bidwell et al. 2013). At the same time, the organizational channels through which markets constrain firm pay-setting remain poorly understood. Economics research emphasizes differences in firm productivity and labor market power (Berger et al. 2024; Card et al. 2018), but provides less insight into how these constraints are intertwined with firms' broader strategic choices (Adner, Csaszar and Zemsky 2014; Rahmandad and Ton 2020).

This study engages with organizational theories of jobs and tasks (Chan and Anteby 2016; Cohen 2013; Wilmers 2020) to examine how the income composition of the available consumer pool constrains firms' pay-setting choices and thereby contributes to wage inequality across firms. Firms position their offerings around the quality preferences of the consumers they target within the market available to them. Because these preferences are strongly shaped by income, higher-income consumers tend to demand higher-quality goods and services and are willing to pay a premium for them (Auer et al. 2024; Dai, Jin and Zou 2025). In response, firms targeting wealthier consumers can design jobs around distinct task sets that support higher service standards (Boyle and Vandebroeck 2025; Sherman 2011). As labor market competition raises the cost of the skills required to perform these tasks, delivering quality may incentivize firms to adopt more specialized job structures to assign the right tasks to the right workers (Kohlhepp 2024). In contrast, firms serving lower-income consumers place greater emphasis on minimizing turnover-related costs by assigning workers a wider range of general tasks (Atencio-De-Leon, Lee and Macaluso 2025). Overall, these differences in job design require firms to hire workers with different skill profiles, contributing to pay differences across firms.

For example, low-end and high-end restaurants both perform core functions such as customer service and table maintenance, but they organize these functions into different jobs and task structures in response to distinct customer demands. In lower-end establishments, a single server typically performs a broad set of simpler duties, from greeting customers to setting tables. High-end restaurants, by contrast, divide these functions into specialized roles. Front servers focus on refined customer service, including personalized menu explanations and wine recommendations, while back servers handle table maintenance under strict protocols that reflect service standards rarely found in lower-end establishments. These differentiated tasks command premium pay to attract more skilled workers, and the resulting labor costs can be passed on to affluent consumers through higher prices. Moreover, when workers capable of performing multiple high-level tasks are scarce, specialization becomes more efficient: workers with strong customer-interaction skills can focus on their comparative advantage rather than ancillary tasks such as table setup, while other workers perform back-server duties that require protocol adherence but less interpersonal skill. In contrast, lower-end restaurants face less customer demand for complex tasks, so most workers can perform the tasks the establishment requires. Employers therefore assign broader responsibilities to each worker, helping mitigate operational losses from high turnover.

I utilize multiple data sources to empirically test this theory. First, I construct a unique establishment-level dataset that captures both the income levels of consumers and the pay levels of employees, with a primary focus on customer-facing service industries. I build this dataset by linking restricted-use administrative records of average workplace pay from the Bureau of Labor Statistics with mobile location data on consumer visits. To the best of my knowledge, this is the first large-scale dataset to jointly measure these two variables at the firm level, allowing for a direct examination of the relationship between consumer income and worker pay.

The correlational analysis shows that market segmentation by consumer income is meaningfully associated with between-firm pay inequality. This pattern is consistent with the expectation that firms build their pay practices around the consumers they target and, on average, succeed in attracting those consumers. In my most restrictive model, which controls for neighborhood and narrow industry fixed effects, a 10% increase in consumer income is associated with a statistically significant 0.6% increase in average establishment pay. Because this comparison is made among establishments within the same industry and neighborhood that serve different consumer bases, it effectively rules out mechanical geographic correlations between consumer income and wages that stem from local cost-of-living differences or variation in the potential customer pool – factors that have long complicated efforts to identify the firm-level link between wages and product market segments (Batt et al. 2020). Additional robustness checks confirm that this relationship is not driven by tips or other forms of extra pay. Moreover, I find evidence that wage differences across firms are shaped in part by within-establishment adjustments: as the income level of the customer base rises, firms tend to raise pay as well. Tracking individual establishments over time, I observe that a 10% increase in consumer income corresponds to an approximate 0.1% rise in pay levels. The predictive power of consumer income on pay is strongest in the accommodation and restaurant industries, where product markets are highly segmented by service quality or status concerns (Baum and Haveman 1997; Favaron, Di Stefano and Durand 2022), and among low-wage, frontline jobs that play an important role in delivering quality (Oliva and Sterman 2001; Sherman 2011).

To probe whether this relationship is causal, I exploit the pandemic-era collapse in commuting, which abruptly removed a large share of commuter visits from commuter-dependent establishments for reasons unrelated to their own pay-setting or business strategies. This setting allows me to test whether losing the particular consumer base around which an establishment had structured its jobs and pay policies induced it to adjust pay. Using each establishment's pre-pandemic commuter share as its exposure to the shock, I estimate event studies that compare establishments within the same narrow industry and local market whose overall customer traffic followed similar trajectories. The interpretation rests on two facts measured directly in the visitor data: exposed establishments experienced large declines in commuter visits, and the commuters who withdrew were, on average, disproportionately affluent, shifting the income composition of exposed establishments' customer bases downward. In the pooled sample, relative pay at exposed establishments fell by roughly 2% per standard deviation of exposure beginning in late 2020 and did not recover even as the measured customer mix rebounded. Disaggregating the results shows that the response was confined to where the withdrawal actually changed the composition of demand: pay fell only in accommodation and food services, the sector in which the customer-income mix meaningfully deteriorated. Within that sector, I find suggestive evidence that the decline was concentrated among establishments that had served the most affluent commuter clienteles. The pattern is consistent with the theory that pay levels are anchored to the consumer base around which an establishment has built its jobs and business model.

To investigate the mechanisms behind the relationship between consumer income and worker pay, I began by conducting informant interviews with restaurant managers across the Greater Boston area.

These interviews highlighted differences in job design between establishments serving higher- versus lower-income consumer bases. To formally test the insights generated from the interviews, I then turned to an online job postings dataset that includes detailed information on task requirements for each role. Comparing postings from firms serving higher- versus lower-income consumers reveals meaningful differences in job design within the same occupation. Tasks at higher-income-serving firms are, on average, associated with higher labor market prices – driven largely by differences in tasks involving customer interactions and those characteristic of frontline roles. These firms also tend to structure jobs around a narrower set of distinct tasks. I supplement this analysis with data from an online pay-review platform, which allows me to control for unobserved, time-invariant individual heterogeneity. The results indicate that wage premiums associated with consumer income are primarily attributable to worker selection rather than rent-sharing. The results suggest that differences in job design reflect firms' efforts to effectively deploy workers with different skills, rather than differences in organizational practices applied to a common pool of workers.

This study advances our understanding of between-firm pay inequality by showing how firms' strategies toward their consumer bases shape wage setting and by tracing pay differences across firms to corresponding differences in job design. Importantly, because firms target particular consumer groups within the pool of consumers available to them, that pool both constrains and enables their pay policies. This argument departs from accounts emphasizing firm discretion in wage setting while also extending constraint-based explanations of between-firm pay inequality, which have largely overlooked the consumer base as an important constraint on firms' pay decisions. Examining how firms respond to this constraint also reveals that the resulting wage differences are rooted in how work itself is organized. The contribution is not simply to show that workers performing more specialized or higher-skilled tasks earn more. Rather, I demonstrate that customer-facing service firms serving different consumer bases systematically reorganize the same occupations into substantively different jobs. These findings underscore the central role of consumer-driven job design in generating wage disparities across service-sector firms.

2 Explaining pay disparities between organizations

2.1 Labor Market Power and Organizational Discretion

While it has long been recognized that organizations differ in how much they pay their workers (Abowd, Kramarz and Margolis 1999; Krueger and Summers 1988), recent attention to between-firm pay gaps has been fueled by mounting evidence that much of the rise in wage inequality over the past four decades has occurred between, rather than within, firms (Barth et al. 2016; Gartenberg and Wulf 2020; Song et al. 2019). Research shows that some firms pay wage premiums and employ more skilled workers than others (Card et al. 2018; Song et al. 2019), yet the underlying sources of these differences remain imperfectly understood.

Employment research has shown that firms exercise discretion in wage setting beyond immediate market constraints, whether through bargaining processes (Avent-Holt and Tomaskovic-Devey 2014), internal pay-setting systems (Cobb and Lin 2017; Doeringer and Piore 1971), or broader organizational practices intended to raise productivity and improve job quality (Rahmandad and Ton 2020; Ton 2014). Yet the generalizability of employer discretion in pay setting – and therefore its contribution to between-firm pay inequality – remains debated. Critics argue that such discretion may be limited to settings in which firms earn sufficient rents and owners are willing to absorb the cost of higher wages, while market constraints dominate in most other contexts (Hall and Krueger

2012; Osterman 2018). Customer-facing service industries, for example, are characterized by intense price competition, low unionization, and a predominantly low-skilled workforce, potentially leaving little scope for discretionary pay practices (Osterman 2020).

Moreover, a growing body of research argues that pay-setting practices have become less closely tied to internal organizational processes and more closely linked to external labor markets (Bidwell et al. 2013; Dencker and Fang 2016). This shift helps explain why recent research on between-firm pay inequality has focused increasingly on how firms face different constraints in setting wages. A prominent example is the monopsony framework in labor economics, which explains between-firm pay variation as the joint product of worker sorting across firms and differences in employers' ability to pay workers below their market value, ultimately tracing these patterns to firm-level differences in productivity and labor market power (Berger et al. 2024; Card et al. 2018; Song et al. 2019). Lower-productivity firms do not simply choose to pay lower wages; they do not have jobs that require higher-skilled workers. At the same time, although firms generally seek to limit labor costs, they vary in their ability to do so because their monopsony power differs, allowing some employers to pay workers less than comparable workers could earn elsewhere.

Yet the literature has devoted relatively little attention to how firms' employment decisions intersect with their broader strategic choices. In particular, existing accounts provide only a partial explanation of the conditions that sustain productivity differences across firms and thereby shape worker sorting and employers' wage-setting power. While previous research has examined how technological change and outsourcing contribute to between-firm pay inequality by increasing worker sorting and affecting the distribution of rents (Bergeaud et al. 2024; Drenik et al. 2023), these factors represent only a narrow subset of the interdependent conditions firms face (Feigenbaum and Gross 2024; Rahmandad and Ton 2020). One important omission is product-market positioning, which organizational research has long treated as central to firms' broader configurations of policies and practices (Adner, Csaszar and Zemsky 2014). This literature suggests that decisions in one domain, such as wage setting, are constrained by choices in other strategic domains, such that some pay levels are more compatible than others with a firm's overall configuration (Levinthal 1997; Rahmandad 2019). To the extent that these interdependencies shape firms' pay practices, their limited treatment in the existing literature leaves an important gap in explanations of between-firm pay disparities.

I propose a theory linking disparities in consumer income to differences in job design and pay across firms. Because higher-income consumers demand higher-quality products and services (Auer et al. 2024; Dai, Jin and Zou 2025), firms oriented toward them reorganize work accordingly: they allocate quality-critical, skill-intensive tasks through more specialized job structures that enable the reliable production of higher-quality outputs. These specialized jobs, in turn, require workers with the skills needed to perform those tasks, generating pay differences across firms. This theory extends existing explanations of between-firm pay inequality by identifying variation in consumer income as an understudied condition supporting differences in how firms organize production and compensate workers. It further suggests that observed pay differences reflect constraints on firms' wage-setting practices arising from the consumers available to them.

2.2 Consumer income, Job Design, and Pay levels at Organizations

Empirical research consistently finds that lower-income consumers are more price-sensitive, placing greater emphasis on price than on other product attributes (Auer et al. 2024; Dai, Jin and Zou 2025). It follows, then, that higher-income consumers tend to value non-price attributes – such as customization, service attentiveness, or symbolic value – more heavily (Gottlieb et al. 2023; Wilmers 2017), which I refer to broadly as *quality* throughout this paper. The well-established relationship between consumer income and quality preferences enables some firms to target more affluent customers by emphasizing higher-quality products and services as they seek to position themselves within distinct market niches (Adner, Csaszar and Zemsky 2014; Podolny 1993; White 1981). An important scope condition follows directly from this argument: the availability of consumers from a given income group ultimately constrains or enables firms' ability to target that group.

How, specifically, do firms supply quality when they target the richer consumers available in the market? Organizational research shows that employers structure jobs and allocate tasks to serve various goals – from cost control to expedient response to external regulations – resulting in substantial variation in task content even among similar roles (Chan and Anteby 2016; Wilmers 2020). Building on this insight, I argue that firms serving higher-income consumers assign workers distinct tasks in order to meet elevated quality expectations. These tasks may involve performing similar functions to those found in lower-end firms but to a higher standard, or they may entail completely different responsibilities that are unique to high-end services. For example, workers at grocery stores serving high-income customers engage in tasks such as custom-cutting meats and operating scratch bakeries, whereas lower-end establishments place less emphasis on customer interaction (Carré and Tilly 2020). Likewise, call center workers in higher value-added segments maintain quality by relying less on scripted responses and dedicating more time to each customer (Batt 2000). In these instances, employers recognize that these more advanced tasks are essential for producing what they describe as "leading-edge goods and services (Carré and Tilly 2020)."

Moreover, differences in job design may extend beyond a single task or role within an organization. When the quality of each task contributes critically to the overall quality of the final product or service

(Kremer 1993), employers may find it advantageous to design jobs such that each worker performs a more advanced set of tasks aligned with the expectations of high-income consumers, thereby raising skill requirements across the organization. For example, in a high-end restaurant, employers do not focus solely on ensuring the food is exceptional; they also emphasize flawless customer interactions and cleanliness to deliver a cohesive final product: the overall dining experience.

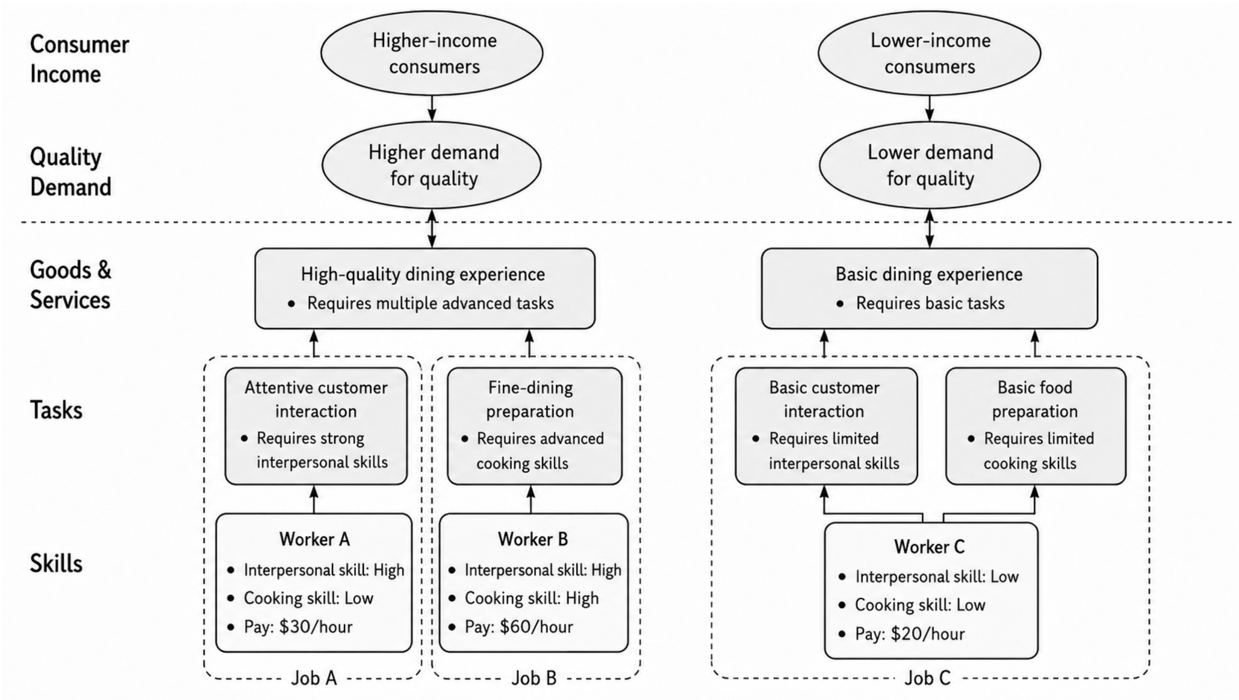
As firms require the performance of higher-skill tasks to serve quality-sensitive consumers, they will increasingly hire workers with the skills necessary to perform those tasks at the desired level of quality. Because richer consumers are willing to pay for these elevated quality levels (Gottlieb et al. 2023), market competition will, in turn, drive up compensation for the workers who possess those skills. For simplicity, I refer to these as *higher* skills: skills that command greater value in the labor market because they are necessary for performing quality-enhancing tasks. These may include interpersonal skills required for high-quality customer interactions, cognitive skills needed to remember and meticulously follow detailed service protocols, or other capabilities that support the production of higher-quality goods and services.

Importantly, employers can increase job specialization to ensure such an advanced set of tasks is performed effectively, when workers who can perform each of those tasks are relatively scarce. At a baseline, firms may have an incentive to limit specialization to mitigate disruptions from turnover and to maintain workforce flexibility by shifting employees across different tasks (Atencio-De-Leon, Lee and Macaluso 2025). However, this approach becomes inefficient when the effective performance of advanced tasks is critical. In such cases, quality may suffer if a worker well-suited for one task is reassigned to another (Cici, Hendriock and Kempf 2025; Coraggio et al. 2025), while hiring employees who can perform multiple advanced tasks equally well can be excessively costly.[1] As a result, establishments that serve high-income consumers are more likely to adopt specialized job structures, whereas their lower-end counterparts may prioritize flexible staffing strategies as a way to minimize the costs associated with turnover.

In summary, I argue that differences in quality demands generate pay disparities across firms targeting different consumer bases. Firms positioned around higher-income consumers face stronger demands for quality. To meet these demands, they design their jobs around an advanced set of tasks critical to the quality of their offerings. In turn, firms offer wage premiums to attract workers capable of performing these high-value tasks, given the potential for higher revenues from serving affluent customers. When workers who can perform multiple high-value tasks are relatively scarce, such firms are further incentivized to adopt more specialized job designs to efficiently ensure quality while minimizing costs. Figure 1 summarizes the theoretical framework of this paper. On this view, an establishment's pay level is a property of the business model it builds around its targeted consumer base – of the jobs it designs and the skills it hires to serve those consumers – rather than a running response to the income of whoever happens to visit. At the same time, the pool of consumers available in a market constrains or enables the strategies firms can feasibly pursue. In markets with few affluent consumers, for example, firms have limited scope to adopt a high-quality, high-pay business model oriented toward affluent customers. Building on this model, I formulate and test three hypotheses in the following sections.

[1]This argument assumes that workers capable of meeting the demands of high-income consumers across multiple roles are relatively scarce compared to those who excel in specific roles; otherwise, employers may not need to adopt such specialized job designs (Kohlhepp 2024)

Figure 1: Theoretical framework



2.2.1 Consumer income and organizational pay levels

First, I examine whether establishments that build their business models around and serve high-income consumers offer higher pay. While the notion that wage levels could be bound by the specific product market that employers operate in is not new (Commons 1909; Osterman 2018), prior research from different traditions has fallen short of clarifying how consumers' ability to pay and workers' pay levels are structurally linked at the firm level.

Hypothesis 1: Establishments targeting higher-income consumers offer higher pay.

I briefly review three related streams of research that address similar questions but are limited in testing the specific theory I propose. These studies either (1) focus primarily on aggregate inequality at the occupation or industry level, rather than on how firms mediate the link between consumers and workers; (2) are limited in their ability to rule out major alternative explanations; or (3) examine contexts that are difficult to generalize beyond their specific setting.

The first line of research examines how local income inequality among consumers influences wage inequality using large-scale quantitative datasets (Gottlieb et al. 2023; Wilmers 2017). Findings from this literature consistently suggest that greater income inequality translates into greater wage disparities. However, the questions these studies address differ from the central hypothesis of this paper. While they investigate whether income inequality spills over into aggregate wage inequality, they provide limited insight into how firms mediate this relationship. The role of firms in this process is not straightforward: it could be that each firm occupies a market niche defined by the income level of its consumers and compensates workers accordingly, as posited here. Alternatively, a single firm may employ workers with varying pay levels who collectively serve consumers across different income groups. Because these studies rely on industry- or region-level data rather than firm-level observations, they are unable to distinguish between these mechanisms.

Another stream of research consists of case studies within specific industries or firms, which have consistently documented that organizations occupying distinct product market positions compensate their workers differently. In particular, they have found that establishments in higher-end market segments or those with higher price points tend to offer higher wages (Batt et al. 2020; Carré and Tilly 2020; Talbert and Bose 1977). Yet much of this work provides little causal evidence and relies on speculative or self-reported classifications of market segments or on price-based proxies (Batt et al. 2020; Carré and Tilly 2020), rather than directly observing variation in firms' consumer bases at scale. This limits their theoretical implications, particularly their ability to assess how the income distribution of the available consumer pool constrains firms' pay-setting strategies.

Additionally, these studies often compare establishments cross-sectionally within broad geographic regions (Batt et al. 2020), making it difficult to disentangle wage differences driven by market segmentation from the regional correlation between income and wages. This issue is particularly pronounced in the United States, where high levels of neighborhood segregation contribute to substantial regional differences in consumer characteristics, even within the same metropolitan area (Berkes and Gaetani 2023; Boar and Giannone 2023). For example, firms located in wealthier neighborhoods may pay higher wages simply because the local cost of living is higher, regardless of their product market segment.

Lastly, research in the trade literature provides several firm-level studies linking product market demand to worker pay. For example, Frías et al. (2024) show that Mexican workers experienced wage increases when their firms expanded exports – primarily to the U.S. – following the 1994 trade liberalization. Similarly, Ma, Muendler and Nakab (2025) find that workers at firms exporting to wealthier countries exhibit steeper wage growth profiles than those exporting to lower-income markets, likely due to differences in human capital accumulation. However, their focus on international trade contexts limits their applicability to understanding how domestic consumer income levels shape worker pay within a national market.

In this study, I examine whether the availability of consumers at different income levels in domestic product markets, and firms' serving of those consumers, influence pay levels within and across firms, thereby addressing the limitations of prior research. I directly measure the income levels of each establishment's consumers to test whether consumer bases constrain and enable establishment-level pay practices, while accounting for geographic confounders.

2.2.2 Consumer income and job design

Next, while prior research has established an initial link between firms' quality orientations and worker pay, the specific organizational mechanisms connecting the two remain less well understood. To explore such mechanisms in greater depth, I conducted informant interviews with restaurant owners and managers in the Greater Boston area – a consumer-facing service industry characterized by pronounced quality differentiation (Favaron, Di Stefano and Durand 2022) and where prior research has documented a strong association between market segments and pay levels (Batt et al. 2020). Although these interviews were conducted in the restaurant sector, they serve to motivate a more general mechanism applicable to consumer-facing service firms that compete on quality differences. The interviews revealed differences in job design across establishments targeting different consumer segments, which informed the development of two hypotheses linking consumer income to job design in the service sector. Further details about the interviews are provided in Appendix I.

Organizational research has shown that the tasks comprising a job are not fixed (Cohen 2013; Wrzesniewski and Dutton 2001), and that significant heterogeneity exists in task composition even among otherwise similar jobs (Feldberg 2022; Martin-Caughey 2021). Existing studies highlight how job design, specifically through the allocation and recombination of tasks, is shaped by managerial responses to cost pressures (Wilmers 2020), organizational uncertainty (Miner 1987; Tan 2015), technological change (Beane 2023), and institutional environments (Lounsbury 2001).

This study introduces consumers – specifically their income levels – as an important but underexamined source of variation in task allocation across employers, with implications for pay differences. While prior research has extensively explored how customers and audiences shape professional jurisdictions (Bourmault and Anteby 2023) and worker identities (Chu 2024; DiBenigno 2022), relatively little attention has been paid to how consumers influence the design of jobs themselves. A few industry-level case studies suggest that firms serving different market segments assign distinct tasks to their employees (Batt 2000; Kelley 1990), but these studies tend to focus on specific task differences rather than on broader patterns in job structure – making it difficult to generalize their findings across sectors or occupations.

I theorize that firms building their businesses around higher-income consumers are more likely to design jobs that include tasks aimed at delivering the higher quality these consumers expect. Such quality-enhancing tasks often involve intensive customer interaction and personalized service (Batt 2000; Carré and Tilly 2020) – for example, "*in a fine dining restaurant, a server may spend 30-plus minutes tableside, talking about food, wine, the menu, and what's on the plate, and just having a conversation*" (Interviewee E). Alternatively, quality may be conveyed through tasks affecting the perceived value rather than functional performance, as in forms of "labor of distinction" (Boyle and Vandebroeck 2025) that work through status mechanisms. In other cases, task variation can reflect heightened precision in executing similar duties – for instance, through formalized protocols that specify exact service standards. As one owner of a higher-end restaurant put it, "*We have an SOP [standard operating procedure] for how to polish the tea cups, the teapot—we have an SOP for every detail*" required to "*consistently deliver the standard we set for our guests*" (Interviewee C). Regardless of the specific channel through which certain tasks contribute to the distinctive qualities valued by affluent consumers, workers with the skills to perform these tasks can command higher wages, as these consumers are willing to pay a premium for them.

> **Hypothesis 2**: Jobs at establishments targeting higher-income consumers comprise tasks that command greater value in the labor market.

In addition, I theorize that job design responses to consumer demand encompass broader shifts in task allocation and organizational structure. Specifically, when workers who can perform multiple distinct tasks at consistently high standards are relatively scarce compared to those who excel at a single task, firms face strong incentives to increase job specialization in order to sustain quality. Although greater specialization may reduce the operational flexibility associated with broader job scopes (Atencio-De-Leon, Lee and Macaluso 2025), it enables firms to control labor costs while meeting elevated quality standards.

As one informant who oversees multiple restaurants across market segments explained: "*(In our high-end restaurant), we don't expect our servers or bartenders to participate in [polishing silverware] because they should be out with guests and selling*" (Interviewee E). This contrasts sharply with the more fluid job structure described by a manager at a casual restaurant: "*Only the general managers or those hired externally have set roles. Otherwise, staff here can easily jump between roles... If I'm bartending and something needs to be done management-wise, I'll do it. If I'm managing and they need help behind the bar, I'll do that too. Everyone just jumps in and helps each other out*" (Interviewee A). The same informant from the first quote further emphasized how job structures differ by market segment: "*[The restaurants have] different hierarchies... largely due to the increasing complexity of service at finer-dining vs. casual. This creates more positions [for finer-dining] on any given night of service and helps us delineate who is responsible for what without overloading anyone*" (Interviewee E). Supporting this view, Batt et al. (2020) finds that the proportion of workers performing both front-of-house and back-of-house roles is significantly higher in fast-food restaurants compared to higher-end establishments, highlighting how job specialization aligns with quality-driven market demands.

Prior employment research emphasizes that employers can enhance product quality and workforce productivity by expanding workers' task variety and enriching job content, typically at the cost of higher pay (MacDuffie 1995; Rahmandad and Ton 2020). I argue, however, that firms serving quality-sensitive consumers do not, on average, combine higher pay with richer task assignments. At the same time, organizational research has long documented the productivity gains of Taylorist forms of work organization, including their capacity to facilitate learning (Boh, Slaughter and Espinosa 2007; Narayanan, Balasubramanian and Swaminathan 2009) and reduce cognitive load (Gong and Png 2023; Staats and Gino 2012). Yet this research has paid less attention to why such systems become more or less prevalent under particular market conditions. Existing studies have largely attributed variation in specialization to inherent tradeoffs between the productivity benefits of specialization and task variety (Narayanan, Balasubramanian and Swaminathan 2009; Staats and Gino 2012), or to internal organizational factors such as coordination costs (Kohlhepp 2024). In contrast, I argue that employers serving quality-sensitive consumers have strong incentives to specialize jobs under labor market constraints. When workers capable of performing multiple high-level tasks are scarce, firms can reorganize work to ensure the reliable execution of critical tasks while limiting labor costs.

Hypothesis 3: Jobs at establishments targeting higher-income consumers comprise more specialized sets of tasks.

3 Predicting organizational pay levels with customer income

In this section, I outline the empirical strategies used to test my first hypothesis on the firm-level relationship between consumer income and pay levels and present the results of the analyses. To do so, I merge a mobile location dataset with administrative records on establishment-level employment and payroll to create a novel dataset that captures both consumer income levels and pay outcomes at the establishment level. This linkage enables the first large-scale analysis of the relationship between the income levels of the consumers around whom firms position their businesses and the pay levels they offer, shedding light on an understudied source of between-firm wage inequality and on the role firms play in transmitting income inequality into pay inequality.

3.1 Data

3.1.1 Consumer Income

The independent variable that will be used throughout this research, consumer income, is derived from the SafeGraph Patterns dataset. SafeGraph collects GPS-based mobile location data and provides monthly counts of unique visitors to each place of interest (POI), which primarily includes customer-facing businesses. The dataset covers the period from the first quarter of 2018 through the second quarter of 2022, which defines the temporal scope of the analysis. Because SafeGraph's coverage is relatively limited before 2020 and the years 2020-2021 were affected by the COVID-19 pandemic, I also replicate the main analysis using only the first two quarters of 2022 as a robustness check, reported in Appendix C.

In the SafeGraph dataset, visitor counts are aggregated by home census block group (CBG), enabling me to link each POI to the demographic and economic characteristics of its visitors' neighborhoods using 2016–2020 American Community Survey (ACS) data. I compute the average visitor income for each POI as a weighted average of CBG-level incomes, using visitor counts from each CBG as weights. Since ACS estimates reflect multi-year averages, CBG-level incomes are treated as fixed over time. However, the composition of visitors to each POI can vary from month to month, allowing for changes in a POI's average visitor income over time. The measure is reasonably reliable, as a CBG represents a small geographic unit, typically 600 to 3,000 residents. However, some within-CBG heterogeneity may still introduce measurement error. To address this concern, I re-estimate my main analysis using only the CBGs with more homogeneous income distributions as a robustness check, as detailed in Appendix C.

This visitor income measure does not exclusively capture *consumers*, as it may also include worker income to some extent. However, I expect the influence of workers to be minimal because each visitor is counted only once per month, regardless of how frequently they visit. Moreover, the average number of visitors in the dataset is significantly larger than the average employment size (see Table 1). Therefore, I anticipate that the variation in visitor income predominantly reflects consumer income.

3.1.2 Wage and Employment

I derive the workplace-level pay variable from the Quarterly Census of Employment and Wages (QCEW), a restricted-access administrative dataset provided by the Bureau of Labor Statistics. The QCEW dataset is sourced from state UI programs and covers all US establishments since 1990. Researchers have access to data from more than half of the states, which defines the scope of my final dataset. The list of included states is provided in Appendix A.

Importantly for this research, the QCEW contains monthly employment and quarterly wage data, allowing me to calculate monthly employment and pay per worker for each establishment.[2] The QCEW employment variable includes part-time and temporary workers, and the wage variable

[2]Note that the QCEW only include establishment-quarter-level wage averages. I supplement these records with a smaller survey that includes establishment-occupation wages in subsubsection 3.3.2.

includes bonuses, stock options, severance pay, profit-sharing, the cash value of meals and lodging, and tips and other gratuities.

I supplement the QCEW pay records with data from Glassdoor, an online platform on which workers report their compensation. Further details on the Glassdoor data and its linkage to SafeGraph are provided in Appendix G. Although the broad QCEW measure of average pay is sufficient for documenting the relationship between consumer income and worker pay, it does not separately identify several potentially important mechanisms, most notably tips and differences in hours worked. Using the Glassdoor pay reports, I find no evidence that the main results are driven by tipping or variation in reported work hours (Appendix M). I also use Glassdoor for several core analyses that could not be conducted with the QCEW because the Bureau of Labor Statistics closed its external researcher access program during the course of this study, making the QCEW microdata unavailable for those analyses. I return to this issue in Section 3.2.

3.1.3 Merge Process

Both SafeGraph and the QCEW datasets include business names and address information, which I use to link workplace-level pay data to consumer incomes. The match rate – comparing the total pool of unique establishments in SafeGraph to those successfully matched – is approximately 36%, with a substantial portion of unmatched cases due to missing address information on the QCEW side. Details on the matching process and associated concerns are provided in Appendix B.

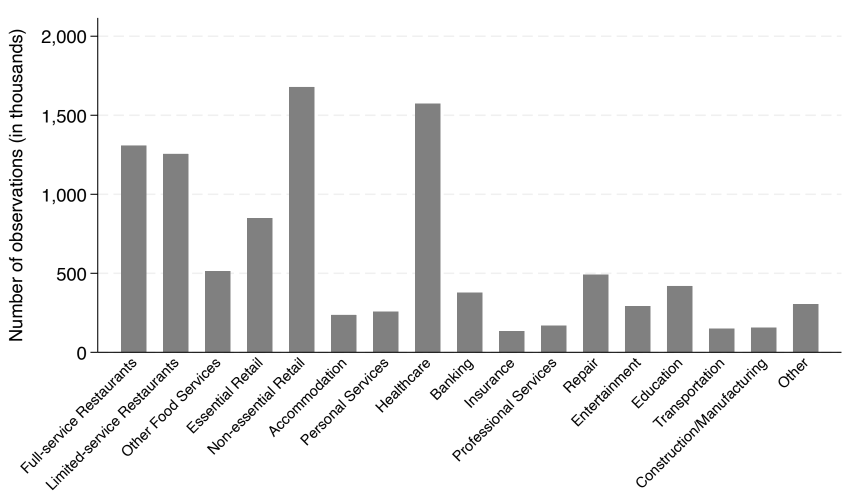
The final matched dataset used in my analysis comprises approximately ten million establishment-by-quarter pairs and around one million unique establishments. Table 1 presents descriptive statistics for this matched sample.

Table 1: Descriptive Statistics of SG-QCEW matched sample (N=9,913,436)

Variable	Mean	S.D.	Min.	Max.
log(Average real pay)	7.63	0.67	-4.00	16.46
log(Median visitor income)	10.98	0.31	7.82	12.42
log(Average employment)	2.31	1.25	-1.09	10.40
Share Black visitors	0.09	0.12	0	1
Share White visitors	0.75	0.17	0	1
Share Hispanic visitors	0.17	0.18	0	1
Share Asian visitors	0.05	0.08	0	1
Share College graduates	0.31	0.13	0	1
Share Less than High school	0.11	0.07	0	1
Share Female	0.50	0.02	0	1
Median Age	39.24	4.58	7.5	86.5
log(Number of visitors)	7.55	1.62	1.81	15.26

* Observations at Time X Workplace level, 2018-2022.

Figure 2: Industry composition of the sample



Note: Data from SafeGraph-QCEW, 2018-2022.

Figure 2 shows the industry composition of the final matched sample, which, as expected, is heavily concentrated in customer-facing sectors, reflecting the underlying structure of the SafeGraph dataset (Massenkoff and Wilmers 2023). Specifically, the sample largely consists of restaurants, retail, and healthcare establishments. However, the overrepresentation of the healthcare sector could be a concern, as consumer income levels may be less relevant for this industry (Osterman 2020). I provide additional analysis excluding healthcare and manufacturing industry observations in Appendix C.

3.2 Methods

If firms build their pay practices around their target consumers, as Hypothesis 1 predicts, and their realized customer bases reflect their intended market positioning on average, we should observe a positive establishment-level association between consumer income and worker pay. I first establish this relationship using the matched QCEW–SafeGraph dataset. A key challenge is separating the predictive relationship between consumer income and firm-level pay from geographic confounders, since establishments in higher-income areas tend both to serve wealthier customers and to employ higher-paid workers. The SafeGraph data help address this issue by providing detailed geographic information on both the home CBGs of visitors and the CBGs in which each point of interest (POI) is located. Using this information, I compare establishments within the same neighborhood and narrow industry, thereby ruling out geographic correlations between income and wages that arise from factors other than consumer demand, such as cost-of-living adjustments to pay. This design ensures that comparisons are made among establishments likely to draw from the same pools of customers and potential workers.

Specifically, I fit cross-sectional fixed effects models that control for quarter-by-CBG-by-industry fixed effects, as shown in Equation 1. This specification enables comparisons between establishments such as Dunkin' and Blue Bottle, or Walmart and Costco, that are located in the same neighborhood but draw consumers from neighborhoods with different income levels. For the industry variable, I use 6-digit codes from the North American Industry Classification System (NAICS), which identifies over a thousand distinct industries.

$$PAY_{et} = \beta * INC_{et} + \lambda \mathbf{X}_{et} + Time_t \times CBG_e \times NAICS_e \tag{1}$$

In Equation 1, PAY_{et} and INC_{et} represent the logged average pay of workers and the logged average income of visitors for establishment e at time t, respectively. Time is defined at the year-by-quarter level. $\mathbf{X}_{et}$ includes establishment-level controls such as the number of employees, the number of visitors, and the sociodemographic characteristics of the visitors.

I include controls for the number of employees and the number of visitors, as these variables may be correlated with wage levels through firm-size wage effects or overall revenue, and may also systematically relate to the income levels of consumers. For instance, although situated in a manufacturing context, prior research finds that larger firms tend to attract higher-income consumers (Faber and Fally 2022). Additional sociodemographic controls – such as age, race, gender, and education – are included in my most restrictive model given their well-documented associations with both income and consumption patterns (Jaravel 2019).

To test whether firms adjust the consumers they target – and their corresponding pay practices – over time in response to the consumer pool available to them, rather than the cross-sectional relationship between consumer income and worker pay merely reflecting persistent strategic differences across firms, I estimate a panel model that slightly modifies Equation 1.

$$PAY_{et} = \beta * INC_{et} + \lambda \mathbf{X}_{et} + Time_t \times CBG_e \times NAICS_e + EST_e \quad (2)$$

Equation 2 incorporates workplace-specific fixed effects (EST_e), enabling within-establishment comparisons of worker pay and visitor income while accounting for temporal trends at the CBG × industry level. However, it is important to note that variation in visitor income over time within establishment is driven solely by compositional shifts in visitors' home CBGs, rather than reflecting changes in CBG-level income over time, as the income variable is derived from the 2016-2020 combined ACS.

Yet while these models can establish whether firms adopt divergent pay practices that align with their respective consumer bases, they cannot determine whether changes in the availability of high-income consumers alter those practices. To strengthen the causal interpretation of the relationship between consumer income and worker pay, I exploit the pandemic-era collapse in commuting to examine how the loss of affluent customers from an establishment's consumer base affects the pay it offers. As commuting fell sharply in 2020 and remained depressed with the persistence of remote work, establishments that had previously depended on commuters lost a substantial share of their customer traffic for reasons unrelated to their own pay-setting or business strategies. Moreover, the visits that were lost came disproportionately from higher-income commuters – plausibly because higher-paying jobs are more amenable to remote work. The logic of the paper's framework in Section 2.2) – in which job design, pay, and market position are chosen jointly around a targeted consumer base – frames the exercise: if establishments build jobs and pay around the consumer base they target – hiring and paying for the skills that affluent customers price – then pay should respond to a durable withdrawal of that base rather than to passing movements in the measured mix.

Although the cross-sectional and panel analyses use QCEW microdata, the Bureau of Labor Statistics closed its external researcher access program in 2025, before I conducted this additional analysis, and has not announced a reopening date. I therefore use pay data from Glassdoor, an online platform on which users report compensation associated with a specific employer, city, and time period. The unit of observation in this analysis is thus firm × city × time, with more than 75% of observations corresponding to a single establishment rather than an aggregation of multiple establishments, thus I refer to them as establishments below for simplicity. Details on the dataset and its linkage to SafeGraph are provided in Appendix G.

I measure each establishment's exposure to commuter shock by its *commuter share* in 2019 – the fraction of its visitors residing outside its own ZIP code – fixed before the pandemic and standardized across firms. Specifically, I estimate

$$\ln w_{iet} = EST_e + \sum_{\tau \neq \text{base}} \beta_\tau \left(\text{Commuter}_e \times \mathbf{1}[t \in \tau]\right) \tag{3}$$

$$+ Time_t \times Geography_e \times NAICS_e + Time_t \times \Delta Traffic_e + \mu \mathbf{X}'_{iet} \tag{4}$$

where w_{iet} is the hourly pay reported by review i at establishment e in calendar time t. The establishment fixed effect EST_e compares each establishment with itself over time, differencing out its permanent characteristics and pay level, and $Time_t \times Geography_e \times NAICS_e$ removes shocks common to a geography, 6-digit industry, and calendar time. Forces such as dining restrictions, industry-wide labor shortages, or shifts in tipping norms are absorbed here, so identification comes from comparing establishments in the same narrow industry, market, and time unit whose dependence on commuters differed.

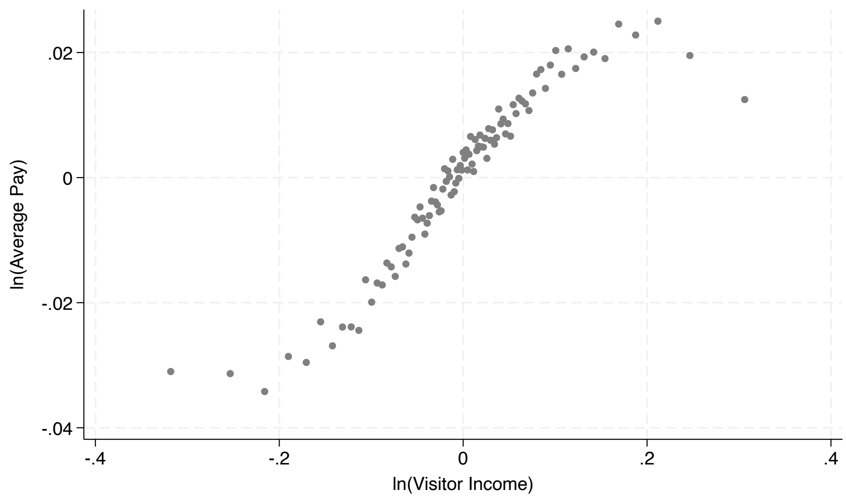
Because the most granular geographic information available in Glassdoor is the city and the data become relatively sparse at that level, I report two specifications that differ only in geographic aggregation and the definition of the event-time bins, τ. In the first, $Geography_e$ is the employer's city and τ runs over calendar years (base 2019), because city-by-industry cells hold too few contemporaneous pay reports to trace a quarterly path with precision; in the second, $Geography_e$ is the metropolitan area and τ runs over calendar quarters (base 2019q4), which affords the timing.

The fixed effects $Time_t \times \Delta Traffic_e$ interact the calendar time with the employer's quintile of realized 2019-2021 change in visitor counts, ranked among establishments in the same 6-digit industry and metropolitan area. Establishments are therefore compared only with peers whose customer traffic followed similar trajectories. The robustness checks in Appendix H show that the results remain largely similar when these controls are omitted. Together with the contemporaneous log visitor-count control in $\mathbf{X'}_{iet}$, these fixed effects hold the volume of customer traffic constant, allowing the estimates to isolate changes in the *composition* rather than the *volume* of demand. $\mathbf{X'}_{iet}$ also includes the worker's reported experience, entered quadratically with a missing-value indicator, and employment status. These controls account for mechanical changes in pay that could arise if the commuter shock also makes workers' commutes more difficult, shrinking the available labor pool and inducing establishments to hire less-skilled workers. Standard errors are clustered two ways, by employer and by metropolitan area. The first-stage panels estimate the same equation after replacing $\ln w$ with average log visitor income and omitting the worker-level controls.

3.3 Results

3.3.1 Does Visitor Income Predict Establishment Pay Levels?

Figure 3: Binned Scatterplot of Establishment Pay and Visitor Income, Adjusted for Time, Neighborhood, and Industry Effects



Note: Data from SafeGraph-QCEW, 2018-2022. Controlling for Time X CBG X NAICS. QCEW average pay is deflated with 2017 dollars. Visitor Income predicted with 2016-2020 combined ACS.

Figure 3 presents a binned scatterplot depicting the relationship between consumer income and worker pay levels. Each dot represents 1% of the visitor income distribution, residualized at the $Time \times CBG \times NAICS$ level. The plot reveals a clear upward-sloping linear trend, except at the very ends of the distribution, indicating that pay levels generally increase with visitor income across establishments.

Building on the graphical evidence presented in Figure 3, I next test whether the observed relationship holds under formal regression analyses with varying levels of control, reported in Table 2. Column (3) presents the coefficient from a specification that corresponds to the Figure 3, including fixed effects for time × CBG × industry. For comparison, Column (1) reports the unadjusted

Table 2: Cross-sectional Pay Regressions on Visitor Income

	(1)	(2)	(3)	(4)
Visitor Income	0.227***	0.295***	0.101***	0.064***
	(0.001)	(0.001)	(0.002)	(0.004)
Fixed effects:				
Time X NAICS		×		
Time X NAICS X CBG			×	×
Controls				×
N	9,907,238	9,906,503	3,632,239	3,395,868

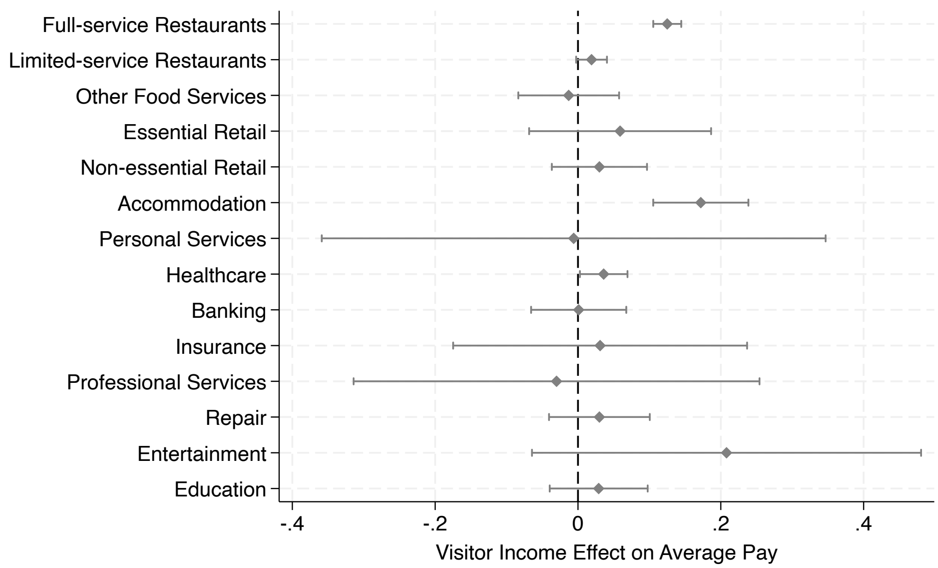
* Observations at Time X Workplace level, 2018-2022. Dependent variables for all models are logged average pay at establishment level from QCEW. Pay deflated with 2017 dollars. Visitor income is median visitor income predicted from SafeGraph with 2016-2020 combined ACS. Time is at year-by-quarter-level. Controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

relationship with no fixed effects, while Column (2) includes time × industry fixed effects but omits CBG fixed effects.

The substantial decline in the coefficient from Column (2) to Column (3) indicates that approximately two-thirds of the cross-sectional correlation between visitor income and worker pay can be explained by the fact that high-income-serving establishments are disproportionately located in high-income neighborhoods. This implies that prior studies linking market segments to pay levels (*e.g.*, Batt et al. 2020) may have overstated the strength of this relationship by not including fine-grained geographic controls. The number of observations in Columns (3) and (4) drops to roughly one-third of the sample, which is expected, as including CBG fixed effects requires multiple establishments within the same narrow industry to be located in the same neighborhood.

Across all specifications, higher visitor income is consistently associated with higher average pay, lending support to the hypothesis that market segmentation by consumer income contributes to between-establishment pay differences. The most saturated model in Column (4), which follows from Equation 1, implies that a 10% increase in visitor income is associated with a 0.6% increase in worker pay. At the sample mean, this estimate suggests that a one-standard-deviation increase in visitor income corresponds to roughly $41 more in monthly pay per worker. Additional robustness checks in Appendix M suggest that this relationship is unlikely to be driven by extra pay, including tips, or by differences in work hours.

Figure 4: Coefficients by industry



Note: Data from SafeGraph-QCEW, 2018-2022. This figure illustrates the results from replicating the analysis in Column (4) of Table 2 by each industry sample.

Figure 4 breaks down the analysis from Column (4) of Table 2 by industry. The findings suggest that the overall positive relationship between visitor income and worker pay is primarily driven by restaurants and accommodation industries, which are known for high market segmentation based on quality or status differentiation at the establishment level (Baum and Haveman 1997; Favaron, Di Stefano and Durand 2022). This pattern is consistent with my theory that differences in customer preferences for quality may help explain how income inequality contributes to disparities in worker pay.

Does the relationship between consumer income and worker pay levels only hold across different establishments? If so, unobserved establishment-level factors – rather than variation in consumer income – could be driving the results, even after controlling for narrowly defined industry and geographic cells. To address this concern, I estimate a panel regression model using the matched SafeGraph-QCEW sample, controlling for time-invariant establishment characteristics as well as industry-by-neighborhood characteristics that vary over time.

Table 3: Panel Pay Regressions on Visitor Income

	(1)	(2)	(3)
Visitor Income	0.004***	0.007***	0.009***
	(0.001)	(0.002)	(0.003)
Fixed effects:			
Workplace	×	×	×
Time	×		
Time X NAICS X CBG		×	×
Controls			×
N	9,847,789	3,602,382	3,368,258

* Observations at Time X Workplace level, 2018-2022. Dependent variables for all models are logged average pay at establishment level from QCEW. Pay deflated with 2017 dollars. Visitor income is median visitor income predicted from SafeGraph with 2016-2020 combined ACS. Time is at year-by-quarter-level. Controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees. Standard errors clustered at each establishment level. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

The estimates from Table 3 show that higher visitor income continues to predict higher worker pay within workplaces over time. A 10% increase in visitor income is associated with a statistically significant 0.1% increase in pay, with standard errors clustered at the workplace level. This pattern suggests that firms' pay practices evolve alongside shifts in the income composition of their consumer base. While the effect is substantially smaller than the cross-sectional relationship reported in

Table 2 – where a 10% increase in visitor income corresponds to a 0.6% increase in pay in the preferred specification – it nonetheless indicates a meaningful degree of strategic flexibility.

The gap between the panel and cross-sectional estimates is consistent with two forces. Quarter-to-quarter movements in measured visitor income partly reflect sampling noise, which attenuates the panel coefficient. And under the framework of Section 2.2, much short-run movement in the measured mix is incidental churn in who visits rather than a change in the consumer base an establishment targets – variation to which pay should respond only weakly. Distinguishing a durable change in the availability of a targeted segment from incidental fluctuation requires a large, persistent shock; Section 3.3.3 examines one.

3.3.2 Occupational Analyses

While regressions of establishment-level pay on consumer income from Table 2 show that establishments serving higher-income consumers tend to pay their workers more on average, these models do not account for occupational composition. As a result, the observed pay gaps may reflect differences in the mix of occupations rather than wage differences within occupations. My theory of job design in response to consumer demand for quality does not rule out the possibility of occupational composition differences – establishments serving wealthier consumers may indeed employ a higher share of high-skilled roles, such as managers or first-line supervisors. However, if occupational composition alone accounts for most of the observed pay gap, it would weaken the empirical support for my framework, which also anticipates meaningful within-occupation pay differences. Such within-occupation differences may arise from cross-firm variation in quality expectations for workers with the same job titles, or from the presence of distinct roles, such as front and back servers in fine-dining restaurants, that are not differentiated by standard occupational codes from general servers in casual dining settings.

To distinguish between these two channels – occupational composition versus within-occupation pay differences – I merge in job-level employment and wage data from the Occupational Employment and Wage Statistics (OEWS) into the SafeGraph-QCEW matched workplace data. OEWS, a restricted-access dataset maintained by the Bureau of Labor Statistics, provides granular occupation-level wage and employment information for sampled establishments, spanning back to 1999. I link the two datasets using establishment identifiers common to both OEWS and QCEW. This allows me to extend the analyses in Table 2 by additionally controlling for occupations. Additional details on the OEWS dataset and its integration into this study are provided in Appendix E.

I begin by testing whether the higher pay associated with serving high-income consumers reflects differences in occupational composition or wage variation within occupations, as shown in Table A.4 and Table A.5. Although the limited sample size from this linkage reduces statistical precision – widening standard errors and rendering the correlation between visitor income and wages statistically insignificant in most models – the stability of point estimates before and after controlling for occupation fixed effects in Table A.5 provides suggestive evidence that consumer income is transmitted to worker pay primarily within occupations. Additional details on these analyses are provided in Appendix F.

If the wage gains associated with serving high-income consumers occur within occupations, which specific occupations are most affected by consumer income levels? While my theory of job design does not predict that the mechanism operates exclusively through any particular occupation or task, it is premised on the idea that consumer demand for quality shapes job design and pay levels. If this is the case, we should observe within-occupation wage differences in roles that are known to be important to the production of quality. Conversely, an absence of such variation would call into question the proposed mechanism linking service quality to pay.

Previous research emphasizes that interaction with customers – whether direct or indirect – is a critical component in the production of service quality (Oliva and Sterman 2001; Sherman 2011). For this reason, I expect pay effects of consumer income among frontline occupations, which comprise a large share of the lower-wage workforce (Osterman 2020). To test this, I replicate the cross-sectional regressions of pay on visitor income with time $\times$ industry $\times$ CBG $\times$ occupation fixed effects (Column (4) of Table A.5) across distinct occupational groups. I first examine heterogeneity in the estimated effects across occupational pay terciles, defined using occupation fixed effects from a two-way workplace-occupation pay model following Wilmers and Aeppli (2021), estimated on the 2012 OEWS sample.

The results in Table 4 reveal that the visitor income effect on wages operates primarily through the base category, the lower-paid occupations. Across model specifications, the consumer income effect on wages for lower-paid occupations was consistently larger than for other pay groups.

Next, I examine effect heterogeneity across occupations based on their roles and tasks. I categorized occupations into four groups using their 2-digit SOC codes: managers, professionals, frontline, and residuals as described in Appendix D.

The results from Figure 5 suggest that the wage effect of consumer income is concentrated among frontline workers, those identified as directly involved in delivering service quality in previous studies. This evidence supports my theory that quality differentiation serves as a key mechanism underlying pay differences across workplaces serving consumers with different income levels.

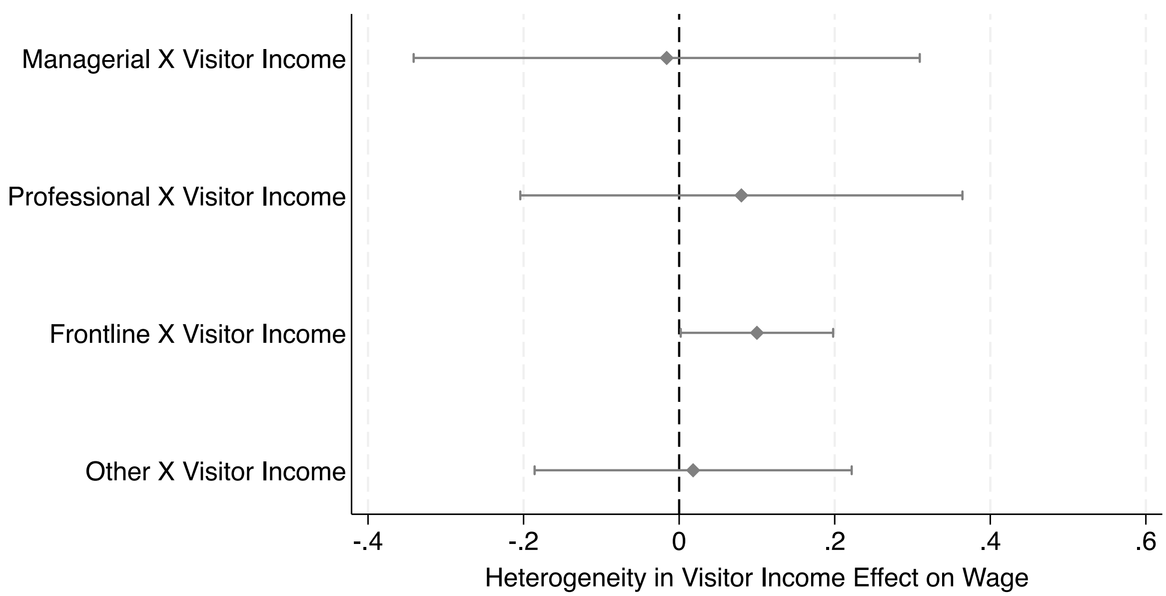
Taken together, the findings from occupational analyses suggest that wage gains associated with high-income consumers are not driven by differences in occupational composition, but rather by

Table 4: Pay Regressions on Visitor Income by Occupation Pay Quantiles

	(1)	(2)
Visitor Income	0.125**	0.209***
	(0.053)	(0.067)
Middle-paying Occs. X Visitor Income	0.010	-0.351***
	(0.062)	(0.103)
High-paying Occs. X Visitor Income	-0.124**	-0.104
	(0.064)	(0.099)
Fixed effects:		
Time X NAICS X CBG X Occ.	×	×
Common Controls	×	
Occs.-Specific Controls		×
N	22,333	22,333

* The regression model in Column (1) lets only the coefficients of visitor income vary for different occupational groups, whereas the model in Column (2) estimates separate coefficients for every variables including the controls. Observations at Time X Workplace X 5-digit Occupation level, matching SafeGraph-QCEW-OEWS 2018-2022. Dependent variables for all models are logged occupational wage predicted from OEWS via midpoint Pareto method following Wilmers and Aeppli 2021. Pay deflated with 2000 dollars. Visitor income is median visitor income predicted from SafeGraph with 2016-2020 combined ACS. Occupation pay quantiles defined from 2012 OEWS, fitting a two-way fixed effects (TWFE) model of pay with 5-digit occupations and workplaces. Time is at year-by-quarter-level. Controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

Figure 5: Effect Heterogeneity by Different Types of Occupations



Note: Data from SafeGraph-QCEW-OEWS matched sample 2018-2022, dropping healthcare and manufacturing industries. Coefficients from fitting the model in Column (4) of Table A.5, separately for each subsample. The four types of occupations add up to the whole sample.

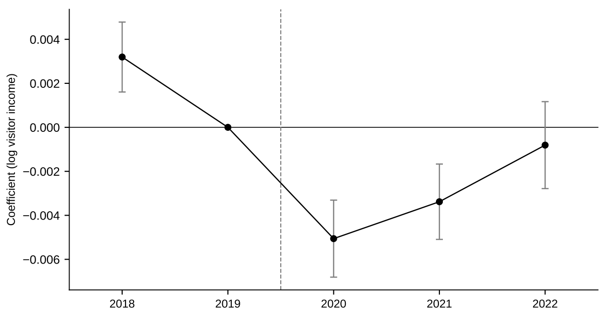
higher payoffs within occupations. Moreover, the effect is especially pronounced in lower-wage, frontline occupations. These patterns are consistent with the theory that pay differences reflect variation in product or service quality and underscore the importance of factors beyond occupational structure in explaining how pay varies across firms serving different consumer segments.

3.3.3 How Does Losing Affluent Consumers Affect Pay?

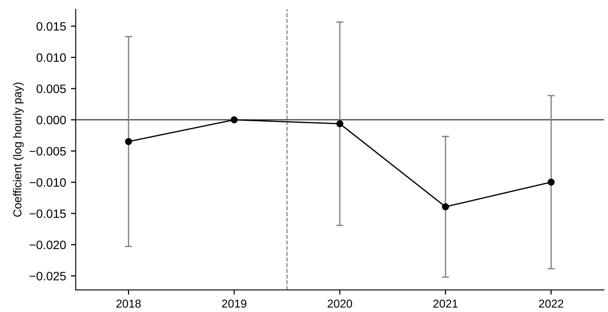
The analyses so far document a robust association between consumer income and establishment pay. Whether that relationship is causal, however, remains an open question. In this section, I use the pandemic-era collapse in commuting to examine whether the loss of affluent consumers led firms to alter their pay policies.

Figure 6 reports the results from estimating Equation 4 for average visitor income and worker pay, pooling establishments across all matched industries. Panels (a) and (c) show the first stage visitor income mix result. Relative to its less-exposed peers, a commuter-dependent establishment's measured visitor income is flat through 2019, falls sharply at the onset of the pandemic – by 1.0% per standard deviation of exposure in the second quarter of 2020, and 0.5% on the year – and recovers to less than a fifth of that decline by the middle of 2022. The first stage is nearly identical when the traffic-quintile comparisons are dropped in Appendix H: the composition of visitors moved almost orthogonally to their number. Because neighborhood incomes are held fixed from the SafeGraph data, this movement reflects a change in *who* visits rather than any change in local incomes. Although the pre-pandemic first-stage coefficients are statistically different from zero given the large sample, their magnitudes are economically negligible and stand in sharp contrast to the steep post-pandemic decline.

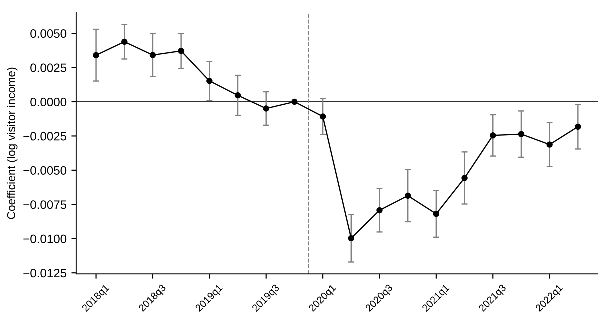
Figure 6: Losing Affluent Commuters and Firm Pay



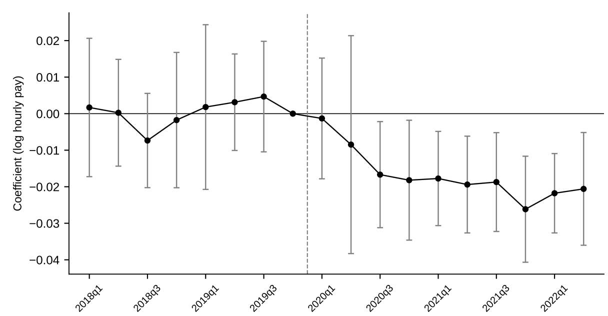
(a) Visitor income, within city, yearly



(b) Hourly pay, within city, yearly



(c) Visitor income, within metro area, quarterly



(d) Hourly pay, within metro area, quarterly

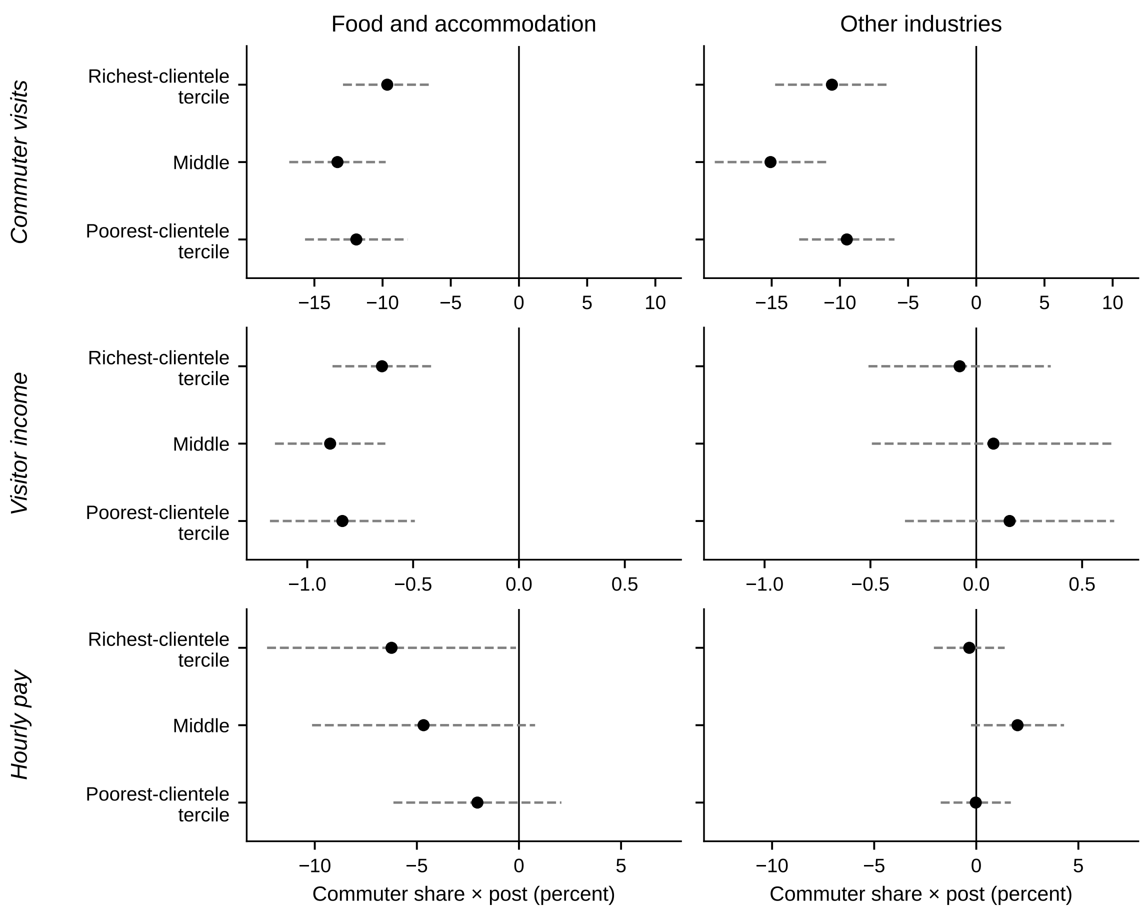
* Event-study coefficients on the establishment's standardized 2019 commuter share. The sample pools all matched industries. The top row interacts commuter exposure with calendar year, with 2019 as the reference period; the bottom row interacts exposure with calendar quarter, with 2019q4 as the reference period. The dashed line marks the onset of the pandemic. All panels include establishment fixed effects and calendar-quarter fixed effects interacted with quintiles of the establishment's realized 2019-2021 change in visitor counts. These quintiles are defined among establishments in the same six-digit industry and metropolitan area, using groups of at least five matched establishments. The top-row panels additionally include quarter×city×industry fixed effects, while the bottom-row panels include quarter×metropolitan-area×industry fixed effects, with industry defined at the six-digit NAICS level. In the left panels, the dependent variable is average log visitor income from SafeGraph for 145,512 matched establishments. In the right panels, the dependent variable is log hourly pay from 662,408 Glassdoor pay reports matched to 98,903 SafeGraph establishments. These models control for reported experience, employment status, and the establishment's contemporaneous log visitor count. Standard errors are clustered by employer and metropolitan area. Bars indicate 95% confidence intervals.

Panels (b) and (d) report the pay response, estimated from contemporaneous Glassdoor pay reports matched to these establishments. Within city and industry (panel b), relative pay at exposed establishments moves nowhere before the pandemic and is still flat in 2020; it falls to 1.4% per standard deviation of exposure below its pre-pandemic level in 2021 and remains 1.0% below in 2022. The quarterly path within metropolitan area and industry (panel d) supplies the timing: nothing through the first half of 2020, a decline that begins in the second half of 2020 – one to two quarters after the customer shock, consistent with the lags in wage adjustment – and deepens to between 2 and 2.6% from late 2020 through the middle of 2022, statistically significant throughout that window. Additional analyses in Appendix H show that these pay declines occurred primarily within occupations: the estimates remain largely unchanged after absorbing occupation-by-time fixed effects, consistent with the cross-sectional findings in Section 3.3.2.

Consistent with my theory, in which firms orient their strategies toward the consumer pool they expect to serve rather than adjust to day-to-day fluctuations in consumer income, pay does not track the recovery of the customer mix. By mid-2022, the measured income composition at exposed establishments had returned to within 0.2% of its pre-pandemic level, yet relative pay remained approximately 2% lower. Unlike routine fluctuations in clientele, the shift to remote work represented a large and salient change that establishments could reasonably have expected to persist. Service models repositioned in response to such a shock are costly to reverse, even after the customer mix begins to recover.

The leading alternative reading of these declines is customer *quantity* rather than customer *composition*. The specification addresses much of this by construction: establishments are compared only against peers in the same industry and metropolitan area whose customer traffic followed the same realized trajectory, with contemporaneous volume controlled as well. What the fixed effects cannot speak to is whether the customers lost differed in ways beyond their number; Figure 7 turns to who they were – and who had built pay around them.

Figure 7: Commuter withdrawal, visitor income, and pay, by industry and clientele income



* Each marker reports the coefficient on the establishment's standardized 2019 commuter share interacted with a post-pandemic indicator (third quarter of 2020 onward), estimated separately by tercile of the baseline income of the establishment's commuter clientele: the median household income of its 2019 visitors residing outside the establishment's own ZIP code, ranked among establishments in the same 6-digit industry and metropolitan area (groups of at least five matched establishments). The left column covers food and accommodation (NAICS 72); the right column all other industries. In the top row the dependent variable is log commuter visits; in the middle row, log visitor income; in the bottom row, log hourly pay from contemporaneous pay reports. The pay row uses the specification of Figure 6. The visitor-income row applies the same specification without worker controls. The commuter-visits row omits the traffic-quintile strata and the visitor-count control, which are constructed from the visit counts themselves. Bars are 95% confidence intervals; standard errors are clustered by employer and by metropolitan area.

Figure 7 provides evidence on *whose* withdrawal altered the composition of demand, and *to whom* that alteration mattered. Establishments are classified by the baseline income of their commuter clientele, measured as the median household income of the home neighborhoods of their 2019 visitors residing outside the establishment's ZIP code. This measure is ranked among establishments in the same 6-digit industry and metropolitan area, ensuring that the classification does not simply reflect differences in industry composition. Within each tercile, I estimate effects on three outcomes: commuter visits, measured visitor income, and hourly pay, with the latter two estimated using the same specification as in Figure 6. The analysis is conducted separately for accommodation and food services – which Figure 4 shows drive the correlational relationship between consumer income and worker pay – and for all other industries, where the estimated effects are largely null.

The results show that commuter visits collapsed across the board: by roughly 10% per standard deviation of exposure in accommodation and food services, and by a similar amount in other industries, across all clientele terciles. The composition of demand, however, did not move in parallel with the decline in visits. Within accommodation and food services, visitor income fell in every tercile – by 0.6 to 0.9%, and if anything by the most at establishments serving the poorest clienteles. Outside these industries, visitor income did not decline in any tercile. This contrast reflects who withdrew. In accommodation and food services, the commuters who continued to visit were measurably poorer than before, indicating disproportionate withdrawal by affluent commuters. In other industries, by contrast, the income of remaining commuters did not fall, while the shift toward local residents working from home made the overall clientele, if anything, more affluent. Pay responded where the composition of demand shifted, not simply where its volume fell: despite similarly large declines in commuter visits across industries, pay fell only in accommodation and food services, where visitor income also declined. This pattern further alleviates the concern that the pay decline in Figure 6 reflects a deterioration in business conditions associated with the commuter shock that is not captured by the controls. The paper's cross-sectional results locate the customer-income pay premium in precisely these industries, and the withdrawal removed the customers on which that premium was premised.

Within food and accommodation, the pay response is detectable only among establishments serving affluent clienteles: statistically significant 6.2% per standard deviation of exposure in the richest-clientele tercile, 4.7% in the middle tercile, and a statistically insignificant 2.0% in the poorest. The size of the measured composition shock cannot account for this pattern: every tercile received the shock, and the richest tercile received, if anything, the smallest. What distinguishes establishments serving affluent clienteles is, as the framework would suggest, the stake: their jobs and pay structures had been built more closely around the customers who withdrew. Yet, it must be noted that this evidence is suggestive, as the difference between the richest- and poorest-clientele responses is not statistically significant, so the data cannot distinguish the erosion of a clientele-specific premium from a uniform, downwardly rigid response to common losing of affluent consumers.

Taken together, the pandemic-era shift strengthens the causal interpretation of the paper's central relationship: when establishments lose the affluent customers around whom their jobs and pay structures were built, their pay falls relative to that of comparable establishments. Moreover, the decline emerges precisely in the industries where the consumer-income pay premium is concentrated and, more suggestively, among the establishments serving the most affluent clienteles. This pattern indicates that wage setting in consumer-facing service establishments is shaped not only by the size of the customer base, but also by the presence – and withdrawal – of affluent consumers.

4 Predicting job design with customer income

This section examines the second and third hypotheses, which were formulated from qualitative insights derived from informant interviews with full-service restaurant managers in the Greater Boston area.[3] These hypotheses propose that jobs in establishments targeting higher-income consumers involve higher-skill tasks and exhibit greater specialization. I formally test these propositions using task-level data from a large job postings dataset.

4.1 Data

The main dataset I use in this section is a matched dataset that I constructed by linking SafeGraph data with job postings from Indeed, collected and processed by Revelio Labs. The job postings, scraped by Revelio Labs starting in 2020, contain employer names and metro regions, which I leverage to link postings to establishments in the SafeGraph dataset. To classify the content of each posting, Revelio Labs applies an internal clustering algorithm that groups tasks based on textual similarity. The level of task granularity is determined by the number of clusters used in the algorithm (e.g., k = 10, 50, 100, 200, 500, 1000, 1500), where a higher k represents a more fine-grained breakdown of tasks. I use these task classifications to measure job specialization, based on the number of distinct tasks associated with each posting. In addition, the job postings dataset includes salary information, which I use to estimate wage premiums associated with particular tasks.

[3] As shown in Figure 4, the full-service restaurant industry is a major contributor to the observed correlation between consumer income and establishment-level pay. Further details on the interview methods, sample, and protocol are provided in Appendix I.

4.1.1 Average Task Price

My theory posits that workers in firms targeting higher-income consumers perform higher-skill tasks that command higher pay. To test this hypothesis, I first infer task-specific market prices by regressing the logged salary of each job posting on binary indicators for the tasks listed in the posting ($k = 1000$ or 1500). This estimation is conducted on a 5% random sample of postings that include salary information, in order to reduce computational burden.[4] Next, I calculate the average labor market value of the tasks associated with each job posting, based on the estimated task prices. For the analysis, I use a standardized measure of average task price to facilitate interpretation of the coefficients. It is worth noting that this method likely yields a conservative measure of task differentiation, as many relevant distinctions (such as subtle differences in quality standards) are unlikely to be fully reflected in job posting descriptions.

4.1.2 Specialization

Following prior research on job specialization (Boh, Slaughter and Espinosa 2007; Gong and Png 2023; Staats and Gino 2012), I measure specialization based on the number of distinct tasks assigned to workers in their roles (*i.e.*, task variety). Greater specialization corresponds to a lower number of tasks performed. To capture this, I construct multiple versions of this measure using different task classifications, each derived from varying numbers of task clusters.

While previous studies highlight an additional dimension of job specialization known as *job turf* – which reflects how distinct a worker's task profile is relative to their coworkers (Wilmers 2020) – I focus solely on task variety to assess specialization within organizations. This choice reflects both

[4] Approximately 11% of 2022 job postings in the Revelio Labs dataset include salary information. This aligns with prior research, which finds that only a small share of online job postings specify salaries – approximately 14% in the case of Batra, Michaud and Mongey (2023).

theoretical and empirical considerations. The relationship between consumer income and job turf is likely to be more ambiguous than that with task variety. For example, employers serving richer consumers may expand staffing for certain specialized roles critical to service quality (e.g., hiring more front servers, thereby reducing job turf), while assigning specific responsibilities exclusively to a smaller subset of workers (e.g., creating host roles dedicated solely to greeting and seating customers, thereby increasing job turf). In addition, I am unable to measure job turf due to data limitations. Unlike prior research that relies on detailed information about internal task allocation within firms (Kohlhepp 2024; Wilmers 2020), my dataset is based on posted job descriptions and provides only partial visibility into actual task distributions.

Table 5 presents descriptive statistics for the core variables used in the analysis. In some models, I include the number of visitors as a control variable to proxy for establishment size,[5] which may be correlated with both consumer segments and task allocation.

Table 5: Descriptive Statistics of Matched SG-Job Postings Sample (N=1,562,393)

Variable	Mean	S.D.	Min.	Max.
log(Salary)	10.37	0.36	5.96	13.07
log(Number of visitors)	8.63	1.41	1.81	15.14
Standardized mean log(Task price) (k=1000)	-0.00	1	-7.05	11.75
Standardized mean log(Task price) (k=1500)	-0.00	1	-8.83	13.30
log(Number of tasks) (k=100)	1.83	0.75	0	3.66
log(Number of tasks) (k=1500)	2.08	0.86	0	4.52
High-income consumers	0.40	0.49	0	1
Share Frontline	0.76	0.42	0	1

* Each observation represents a job posting from 2022 Revelio Labs linked to SafeGraph visitor information. The mean task price for each job is calculated as the average of its associated task prices, where task prices are estimated by regressing salary on binary indicators for each task. Frontline occupations classified based on 2-digit SOC codes, as detailed in Appendix D.

[5]Employment size data is unavailable in both the SafeGraph and job postings datasets.

4.2 Methods

Unlike the SafeGraph dataset, which contains establishment-level identifiers, the job postings dataset includes only firm-level identifiers, and the most detailed geographic information available is at the metropolitan statistical area (MSA) level. Accordingly, I define employers at the firm-by-metro-region level. To construct consumer characteristics at this same level, I aggregate SafeGraph establishment data by averaging visitor income deciles across establishments, controlling for time × industry × metro region cells.[6] To improve the interpretability of these averaged deciles, I then define a binary indicator for a high-income consumer base, equal to 1 when a firm-by-MSA's average income decile exceeds 5.5.

Specifically, for each job posting j listed by firm-by-metro unit f in quarter t, I estimate variants of Equation 5:

$$TASK_{jt} = \beta * HIGHINC_f + \lambda \mathbf{X}_{jt} + Time_t \times MSA_f \times NAICS_f \times SOC_j \tag{5}$$

In Equation 5, $TASK_{jt}$ represents one of two job design outcomes constructed from the posting's task content: the standardized average task price or the logged number of distinct tasks. $HIGHINC_f$ is the indicator for a high-income consumer base defined above, and $\mathbf{X}_{jt}$ includes the logged number of visitors to the firm's SafeGraph establishments in the metro region, proxying for establishment size. The fixed effects compare postings within the same quarter, metropolitan area, 6-digit NAICS industry, and 6-digit SOC occupation, so that β captures differences in how firms serving different

[6] I use income deciles rather than raw consumer income when collapsing establishment-level data to the firm-by-metro level. Deciles are computed from establishment-level visitor income residualized on metro × industry × quarter cells, so that an establishment's value reflects its position relative to its local market rather than regional differences in income levels. Ranking the residuals into deciles then bounds the influence of establishments located in extreme-income neighborhoods when averaging across a firm's establishments (with visitor weights), and places all firms on a common scale.

consumer bases design jobs for the same occupation, posted in the same market at the same time. Baseline specifications omit the occupation interaction and the visitor-count control.

In Appendix K, I report two sets of robustness checks. First, I restrict the sample to firm-by-MSA observations for which the SafeGraph data indicate that the firm operated only one establishment in the MSA, making the match more likely to approximate an establishment-level match. The results remain consistent under this alternative definition of the unit of analysis. Second, I replace the high-income consumer indicator with the continuous average-decile measure and again obtain substantively similar results. Further details on the matching procedure are provided in Appendix J.

For the analysis, I restrict the sample to job postings from year 2022. This is because 2022 is the only year with full data coverage from both sources and is less likely to reflect distortions from the height of the COVID-19 pandemic (SafeGraph data is available through 2022, while Revelio Labs postings are available starting in 2020).

4.3 Results

4.3.1 Consumer Income and Task Prices

I begin by examining whether higher consumer income is associated with workers performing tasks that command greater market value. This analysis again rests on the premise that firms targeting high-income consumers do in fact serve such consumers on average. If these firms design jobs around higher-priced tasks, consumer income should be positively associated with the market value of the tasks included in their job postings.

Table 6 reports results from regressions of the average market value of tasks in each job posting on an indicator for firms serving high-income consumers. Across specifications that vary in

Table 6: Average Task Price Across Jobs

Granularity of work activities	k = 1000		k = 1500	
	(1)	(2)	(3)	(4)
High-income Consumer Base	0.111***	0.084***	0.076***	0.063***
	(0.002)	(0.002)	(0.002)	(0.001)
ln(Number of Visitors)		×		×
Fixed effects:				
Time X MSA X Ind.	×		×	
Time X MSA X Ind. X Occ.		×		×
N	1,290,976	1,248,986	1,290,976	1,248,986

* Observations represent individual job postings from Revelio Labs 2022, excluding postings from the manufacturing and healthcare industries. The dependent variable is the standardized average task price associated with each job, constructed using either k = 1000 or k = 1500 task classifications. This variable captures how jobs are composed of tasks with differing market values. Task prices are estimated by regressing logged salaries on task indicators using a 5% random sample of postings. The dummy variable for high-income consumer base is identified with average visitor income, controlling for Time X Region X NAICS fixed effects with 2022 SafeGraph dataset. Regions at each Metropolitan statistical area. Industry and occupation are measured at the 6-digit NAICS and SOC levels, respectively. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

control variables and task classification schemes, the results consistently support the prediction from Hypothesis 2: firms targeting wealthier clientele place greater emphasis on tasks associated with higher market prices. Importantly, and consistent with the results in Table A.5, this shift toward higher-value tasks is driven primarily by within-occupation variation. Such variation may reflect both differences within nominally identical jobs (*e.g.*, dishwashers in high-end versus casual restaurants) and differentiation across jobs within the same occupation (*e.g.*, front servers in high-end restaurants versus general servers in casual restaurants). Together, these patterns suggest that systematic differences in task composition contribute to the within-occupation pay differentials observed across firms.

Which types of tasks primarily drive the divergence in average task prices across job postings? Similar to the motivation for my occupation-level analysis in Section 3.3.2, while my theory of job design does not predict that the mechanism operates exclusively through any particular task, it is premised on the idea that consumer demand for quality shapes job design and pay levels. A clear implication is that jobs serving higher-income consumers should involve a greater share of tasks pivotal to delivering quality; failure to observe such differentials would weaken the argument that consumer demand for quality links income, job design, and pay.

Relatedly, prior work identifies customer contact – direct or indirect – as central to producing service quality (Oliva and Sterman 2001; Sherman 2011). For this reason, I expect employers serving higher-income consumers to adjust job content particularly in tasks that directly shape the consumer's perception of service quality – those that are characteristic of frontline roles or involve customer interaction. Accordingly, I examine whether increases in task prices are concentrated among frontline or customer-facing tasks.

I identify tasks characteristic of frontline jobs – referred to as *frontline tasks* – using the job postings data. Specifically, I regress a binary indicator for each task on a dummy variable for frontline occupations, as defined in Appendix D, controlling for time × metro area × industry fixed effects. A task is classified as a frontline task if it is significantly more likely to appear in postings for frontline occupations at the 95% confidence level. Tasks not meeting this criterion are designated as non-frontline tasks. Separately, I define *customer-facing tasks* as those whose textual descriptions contain the terms "customer," "guest," or "client." Tasks that do not include any of these terms are classified as non-customer-facing.

Next, I compute the average task price for each job using only frontline, non-frontline, customer-facing, or non-customer-facing tasks. I then replicate the analysis from Column (4) of Table 6, using each of these disaggregated average task-price measures as the dependent variable. This approach enables me to identify which types of tasks primarily drive the differences in average task prices reported in Table 6.

Table 7: Average Task Price Across Jobs, for Specific Task Categories

	Frontline Task	Non-Frontline Task	Customer -facing Task	Non-Customer -facing Task
High-income Consumer Base	0.109***	-0.011***	0.114***	0.007***
	(0.002)	(0.002)	(0.002)	(0.002)
ln(Number of Visitors)	×	×	×	×
Fixed effects:				
Time X MSA X Ind. X Occ.	×	×	×	×
N	1,248,986	1,248,986	1,248,986	1,248,986

* Observations represent individual job postings from Revelio Labs 2022, excluding postings from the manufacturing and healthcare industries.The dependent variable is the standardized average price of specific task types associated with each job, constructed using $k = 1500$ task classifications. This variable captures how jobs are composed of certain type of tasks with differing market values. Frontline tasks are defined as those significantly more likely to be performed by frontline occupations, identified by regressing task indicators on a frontline occupation dummy (see Appendix D for the definition of frontline occupations). Customer-facing tasks are those whose descriptions include the terms "customer," "guest," or "client." Non-frontline (Non-customer-facing) tasks are those that are not classified as frontline (customer-facing). Task prices are estimated by regressing logged salaries on task indicators using a 5% random sample of postings. The dummy variable for high-income consumer base is identified with average visitor income, controlling for Time X Region X NAICS fixed effects with 2022 SafeGraph dataset. Regions at each Metropolitan statistical area. Industry and occupation are measured at the 6-digit NAICS and SOC levels, respectively. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

The results in Table 7 indicate that the increase in task prices associated with serving high-income consumers occurs primarily within task groups classified as frontline or customer-facing. In other words, workers in firms serving high-income consumers perform higher-value frontline or customer-facing tasks than their counterparts in firms serving low-income consumers. By contrast, consumer income has little, if any, effect on the composition of non-frontline or non-customer-facing tasks, suggesting that task upgrading is concentrated in tasks more directly connected to the customer experience.

4.3.2 Consumer Income and Job Specialization

Next, I test whether higher consumer income is associated with greater job specialization, measured by the number of unique tasks associated with each job. This serves as a direct test of Hypothesis 3.

Table 8: Number of Distinct Tasks by Consumer Income

Granularity of tasks	k = 100		k = 1000		k = 1500	
Sample Occ.	(1)	(2)	(3)	(4)	(5)	(6)
High-income Consumer Base	-0.080***	-0.029***	-0.091***	-0.029***	-0.090***	-0.029***
	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.002)
ln(Number of Visitors)		×		×		×
Fixed effects:						
Time X MSA X Ind.	×		×		×	
Time X MSA X Ind. X Occ.		×		×		×
N	1,290,976	1,248,986	1,290,976	1,248,986	1,290,976	1,248,986

* Observations represent individual job postings from Revelio Labs for the year 2022, excluding postings from the manufacturing and healthcare industries. The dependent variable is the logged number of tasks identified from job postings. Results are presented separately based on the task definitions used, clustering job descriptions into either 100, 1000, or 1,500 task categories. The dummy variable for high-income consumer base is identified with average visitor income, controlling for Time X Region X NAICS fixed effects with 2022 SafeGraph dataset. Regions at each Metropolitan statistical area. Industry and occupation are measured at the 6-digit NAICS and SOC levels, respectively.
* Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

Table 8 reports the results from regressions of the logged number of tasks per job posting on an indicator for whether a firm serves high-income consumers. To assess robustness, the analysis is replicated using three different task classification schemes (k = 100, 1000, and 1500). All models include fixed effects for region, industry, and time to ensure comparisons are made within similar contexts. The results consistently show that jobs at firms catering to higher-income consumers involve fewer tasks. As with the findings in Table 6, the results remain robust to the inclusion of occupation fixed effects and controls for establishment size, proxied by the number of visitors.

These findings support the hypothesis that jobs in establishments targeting higher-income consumers tend to be more specialized. Taken together with the results from Table 6, the empirical patterns align with my theoretical prediction that firms may design more specialized jobs to deliver higher-quality services more efficiently. Notably, this pattern contrasts with prior evidence suggesting that firms often improve product quality by enabling workers to perform a broader range of tasks (Ichniowski, Shaw and Prennushi 1997; MacDuffie 1995), implying that firms may adopt different strategies for achieving quality depending on the setting. Indeed, further analysis in Appendix L shows that the pattern documented in this paper is largely driven by the Accommodation and Food Services sector (2-digit NAICS code 72), which has received relatively little attention in existing research on task variety and product quality.

4.3.3 Worker Selection

Although the finding that jobs serving higher-income consumers involve different task content may help explain their higher pay, it remains unclear whether these differences reflect worker selection or rent-sharing mechanisms, such as compensation for greater effort (Cappelli and Neumark 2001; Cappelli and Chauvin 1991). That is, firms may not need to hire more skilled workers and specialize jobs to deploy those skills efficiently; instead, they could assign more demanding tasks to a largely similar pool of workers by offering higher wages.

Ruling out this alternative explanation is central to my theoretical interpretation of what differential job designs across firms targeting different consumer groups represent. If higher pay reflects rent-sharing rather than selection on skill, observed specialization would not constitute a necessary response to skill scarcity. Instead, it could emerge as (1) a mechanical consequence of higher wages increasing worker attachment (Shapiro and Stiglitz 1984), thereby weakening employers' incentives for flexible task assignment (Atencio-De-Leon, Lee and Macaluso 2025), or (2) a means of accelerating learning-by-doing (Narayanan, Balasubramanian and Swaminathan 2009). Both mechanisms provide key alternative explanations to the theory advanced in this study.

To assess whether rent sharing is the primary driver of pay premiums associated with serving high-income consumers, I use a dataset that enables comparisons of pay outcomes for the same individual across different jobs. This allows me to distinguish whether pay differences are due to worker selection – where higher-skilled individuals sort into higher-paying firms – or to rent-sharing practices. Specifically, I match SafeGraph's visitor income data to wage reports from Glassdoor and leverage Glassdoor's anonymized user identifiers. Additional details on the Glassdoor dataset and the matching process are provided in Appendix G.

If pay premiums are primarily driven by firm-level rent sharing rather than differences in worker characteristics, then controlling for individual-specific factors should not substantially change the observed relationship between consumer income and pay. Conversely, a significant reduction in the coefficients would suggest that pay premiums in firms serving higher-income customers largely reflect worker selection effects. To test this, I estimate regressions of reported hourly pay on SafeGraph's high-income consumer indicator, both with and without individual user fixed effects.

The results in Table A.14 suggest that pay premiums at firms serving higher-income consumers are largely attributable to differences in the workers they employ. Controlling for individual worker fixed effects reduces the estimated coefficient on the high-income consumer indicator by 64–72%, depending on whether flexible controls for experience are included. This attenuation challenges the view that pay premiums are primarily driven by rent-sharing and instead points to worker selection as an important contributor, lending support to my argument that employers design specialized jobs to minimize hiring costs in the labor market. The further reduction in the estimated premium when accounting for experience suggests that differences in relevant experience may explain part of the observed pay gap – primarily through selection effects.

5 Discussion and Conclusion

The findings of this paper show that the consumer bases firms target help generate pay disparities between organizations. The pool of consumers available in a market shapes which strategies firms can feasibly pursue, and thereby both constrains and enables their wage-setting practices. Even within the same narrow industry and neighborhood, the income of an establishment's consumers is an important predictor of the pay it offers, and establishments modestly adjust pay as the income composition of their customer base shifts over time. Robustness checks indicate that these results are not driven by differences in work hours or by tips and gratuities (Appendix M). Evidence from the pandemic-era collapse of commuting sharpens the causal interpretation. When commuter-dependent establishments lost the affluent segment of their customer base, their pay declined. Suggestive evidence indicates that this decline was concentrated among establishments whose business models had been built around affluent consumers, consistent with the argument that the available consumer pool disciplines the pay policies firms can sustain.

The results also show that firms targeting high-income consumers design jobs around tasks that command higher market prices, consistent with the premise that affluent consumers are willing to pay for quality and that the tasks producing it are priced accordingly. The task-price differences associated with consumer income are concentrated in customer-facing duties and in tasks characteristic of frontline occupations – precisely where prior research locates the production of service quality (Oliva and Sterman 2001) – supporting a quality-based reading of the relationship. Employers serving higher-income consumers also structure work into more specialized jobs and staff them with workers who command higher pay across employers, in line with my argument that specialization is how firms deploy scarce workers capable of performing quality-critical tasks where they are most effective.

The relationship between consumer income and pay levels is particularly pronounced in industries characterized by substantial quality or status differentiation (Baum and Haveman 1997; Favaron, Di Stefano and Durand 2022), supporting the view that market segmentation is a key driver of wage disparities. Heterogeneity across industries and occupations further reveals an important boundary condition of the proposed theory: the relationship is strongest in frontline occupations and within the restaurant and accommodation industries, where service quality is produced directly through frontline labor and where higher labor costs can be passed on to consumers willing to pay for quality. In other sectors, product differentiation may rely less directly on frontline job design, or quality may be embodied in branding, supply chains, or capital intensity rather than in worker tasks. In these contexts, firms may be able to charge higher prices without reorganizing jobs or raising wages, weakening the link between consumer income and pay. Understanding when and why product market strategies translate into differences in job design and compensation remains an important direction for future research.

At the same time, limitations of the dataset and research design must be acknowledged. The sample is primarily composed of customer-facing service industries due to the characteristics of the SafeGraph dataset. These industries may differ from others in their employment and market strategies (Baum and Haveman 1997; Favaron, Di Stefano and Durand 2022; Osterman 2020), potentially limiting the generalizability of the results. Thus, the primary contribution of this study is to identify a novel mechanism of between-firm pay inequality, rather than to claim that this mechanism operates universally across sectors.

I also use a job postings dataset to examine how differences in job design help explain the link between firms' product-market strategies and pay levels. However, task descriptions in job postings may imperfectly capture the actual tasks performed on the job for various reasons. While it is reasonable to assume that posted task descriptions generally correlate with true job content on average, I cannot fully rule out the possibility that deviations between posted and actual tasks are systematically correlated with consumer income, which could bias the results.

This research identifies an important but underexamined constraint on employers' pay decisions: consumer income. In contrast to perspectives that portray firm-level wage-setting as relatively independent of market forces (Avent-Holt and Tomaskovic-Devey 2014; Ton 2014), the evidence presented here points to structured, rather than purely discretionary, variation in how organizations respond to consumer demand – variation that is ultimately bounded by the pool of consumers available to them. At the same time, this perspective complements prior research on the economic constraints shaping firm pay-setting (Berger et al. 2024; Card et al. 2018). Whereas this literature often abstracts from how employment practices are embedded within firms' broader strategic choices, the findings show that wage variation is systematically aligned with both the characteristics of consumer demand and firms' corresponding choices about job design.

This study also contributes to research on job design by identifying a novel demand-driven mechanism that shapes task assignment. Prior research has shown how employers' efforts to minimize costs (Wilmers 2020), cope with uncertainty (Miner 1987; Tan 2015), and respond to technological change (Beane 2023) can generate variation in job design. Yet, how consumers contribute to the creation of different jobs has received little attention. Conversely, although scholars have increasingly examined how consumers and clients shape organizational practices and worker identities (DiBenigno 2022; Osterman 2020), their influence on task allocation remains underexplored. This study addresses this gap by showing that organizations serving wealthier consumers tend to assign workers more advanced customer-service tasks, suggesting that consumer expectations can indirectly shape job design through managerial efforts to deliver quality.

Lastly, this study contributes to employment research both empirically and theoretically. Empirically, linking consumer foot traffic to worker pay at the establishment level allows me to address questions and potential confounders that prior research has been unable to examine directly. Previous studies have analyzed the relationship between worker pay and either prices or self-reported market segments at the firm level (Osterman 2020; Talbert and Bose 1977), or have measured pay and consumer income at the industry-by-region level (Gottlieb et al. 2023; Wilmers 2017). In contrast, this study measures both worker pay and consumer income at the establishment level on a large scale, making it possible to assess whether the income levels of the consumers an establishment serves are systematically associated with its workers' pay. Moreover, prior research has generally accounted for geography only at relatively broad regional levels (Batt et al. 2020), even though consumer income varies substantially across narrowly defined neighborhoods. This geographic heterogeneity may mechanically generate higher wages through local cost-of-living differences, thereby producing a potentially spurious relationship between consumer income and worker pay. By directly observing establishment-level consumer income, the novel data used in this study enable comparisons among establishments located within the same narrowly defined neighborhoods.

Theoretically, the paper argues that job specialization, rather than task richness, can be bundled with pay premiums to produce higher-quality service. Prior research in manufacturing has emphasized that broader task assignments can improve quality by fostering worker flexibility and cross-functional collaboration (Ichniowski, Shaw and Prennushi 1997; MacDuffie 1995). My findings suggest that this relationship operates differently in service settings. Firms serving higher-income consumers tend to assign workers narrower task sets, even though these consumers are likely to place greater emphasis on service quality. This pattern indicates that the relationship between job specialization and quality is context-dependent. Although the evidence presented in this study is consistent with the argument that firms may adopt specialized jobs to deliver quality when workers capable of performing multiple quality-producing tasks are scarce, future research should clarify the conditions under which quality production becomes coupled with specialized job structures.

The broader implication is that inequality does not end at the point of consumption. Differences in consumers' purchasing power are connected to the kinds of employment systems that firms can sustain, linking segmentation in product markets to stratification in labor markets. Organizations are therefore not merely settings in which consumer inequality is reflected in pay; they are sites where distinctions among consumer markets become embedded in task boundaries, skill requirements, and workers' economic opportunities. The reach of household income extends beyond what households can purchase to the structure and rewards of the work performed on their behalf.

References

Abowd, John M., Francis Kramarz, and David N. Margolis. 1999. "High Wage Workers and High Wage Firms." Econometrica, 67(2): 251–333. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/1468-0262.00020.

Adner, Ron, Felipe A. Csaszar, and Peter B. Zemsky. 2014. "Positioning on a Multiattribute Landscape." Management Science, 60(11): 2794–2815.

Atencio-De-Leon, Andrea, Munseob Lee, and Claudia Macaluso. 2025. "Does Turnover Inhibit Specialization? Evidence from a Skill Survey in Peru." American Economic Review: Insights, 7(1): 56–70.

Auer, Raphael, Ariel Burstein, Sarah Lein, and Jonathan Vogel. 2024. "Unequal Expenditure Switching: Evidence from Switzerland." The Review of Economic Studies, 91(5): 2572–2603.

Avent-Holt, Dustin, and Donald Tomaskovic-Devey. 2014. "A Relational Theory of Earnings Inequality." American Behavioral Scientist, 58(3): 379–399.

Barth, Erling, Alex Bryson, James C. Davis, and Richard Freeman. 2016. "It's Where You Work: Increases in the Dispersion of Earnings across Establishments and Individuals in the United States." Journal of Labor Economics, 34(S2): S67–S97.

Batra, Honey, Amanda Michaud, and Simon Mongey. 2023. "Online Job Posts Contain Very Little Wage Information."

Batt, Rosemary. 2000. "Strategic segmentation in front-line services: matching customers, employees and human resource systems." The International Journal of Human Resource Management, 11(3): 540–561.

Batt, Rosemary, Tashlin Lakhani, Jae Eun Lee, and Can Ouyang. 2020. "The Quality of Jobs in Restaurants." In Creating Good Jobs: An Industry-Based Strategy. The MIT Press.

Baum, Joel A. C., and Heather A. Haveman. 1997. "Love Thy Neighbor? Differentiation and Agglomeration in the Manhattan Hotel Industry, 1898-1990." Administrative Science Quarterly, 42(2): 304–338.

Beane, Matthew. 2023. "Resourcing a Technological Portfolio: How Fairtown Hospital Preserved Results While Degrading Its Older Surgical Robot." Administrative Science Quarterly, 68(3): 691–733.

Bergeaud, Antonin, Clement Malgouyres, Clement Mazet-Sonilhac, and Sara Signorelli. 2024. "Technological Change and Domestic Outsourcing." Journal of Labor Economics.

Berger, David, Kyle Herkenhoff, Andreas R. Kostøl, and Simon Mongey. 2024. "An Anatomy of Monopsony: Search Frictions, Amenities, and Bargaining in Concentrated Markets." NBER Macroeconomics Annual, 38: 1–47.

Berkes, Enrico, and Ruben Gaetani. 2023. "Income Segregation and the Rise of the Knowledge Economy." American Economic Journal: Applied Economics, 15(2): 69–102.

Bidwell, Matthew, Forrest Briscoe, Isabel Fernandez-Mateo, and Adina Sterling. 2013. "The Employment Relationship and Inequality: How and Why Changes in Employment Practices are Reshaping Rewards in Organizations." Academy of Management Annals, 7(1): 61–121.

Boar, Corina, and Elisa Giannone. 2023. "Consumption Segregation."

Boh, Wai Fong, Sandra A. Slaughter, and J. Alberto Espinosa. 2007. "Learning from Experience in Software Development: A Multilevel Analysis." Management Science, 53(8): 1315–1331.

Bourmault, Nishani, and Michel Anteby. 2023. "Rebooting One's Professional Work: The Case of French Anesthesiologists Using Hypnosis." Administrative Science Quarterly, 68(4): 913–955.

Boyle, Bryan, and Dieter Vandebroeck. 2025. "The Labor of Distinction: Butlers, Service Work, and the Production of Elite Lifestyles." American Sociological Review, 00031224241305357.

Cappelli, Peter, and David Neumark. 2001. "Do "High-Performance" Work Practices Improve Establishment-Level Outcomes?" ILR Review, 54(4): 737–775.

Cappelli, Peter, and Keith Chauvin. 1991. "An Interplant Test of the Efficiency Wage Hypothesis." The Quarterly Journal of Economics, 106(3): 769–787.

Card, David, Ana Rute Cardoso, Joerg Heining, and Patrick Kline. 2018. "Firms and Labor Market Inequality: Evidence and Some Theory." Journal of Labor Economics, 36(S1): S13–S70.

Carré, Françoise, and Chris Tilly. 2020. "Retail Jobs: What Pathways for Improvement?" In Creating Good Jobs: An Industry-Based Strategy. The MIT Press.

Chan, Curtis K., and Michel Anteby. 2016. "Task Segregation as a Mechanism for Within-job Inequality: Women and Men of the Transportation Security Administration." Administrative Science Quarterly, 61(2): 184–216.

Chu, James. 2024. "How Beneficiaries Become Sources of Normative Control." Administrative Science Quarterly, 69(2): 324–369.

Cici, Gjergji, Mario Hendriock, and Alexander Kempf. 2025. "Finding Your Calling: Matching Skills with Jobs in the Mutual Fund Industry." Management Science.

Cobb, J. Adam, and Ken-Hou Lin. 2017. "Growing Apart: The Changing Firm-Size Wage Premium and Its Inequality Consequences." Organization Science, 28(3): 429–446.

Cohen, Lisa E. 2013. "Assembling Jobs: A Model of How Tasks Are Bundled Into and Across Jobs." Organization Science, 24(2): 432–454.

Commons, John R. 1909. "American Shoemakers, 1648–1895 A Sketch of Industrial Evolution." The Quarterly Journal of Economics, 24(1): 39–84.

Coraggio, Luca, Marco Pagano, Annalisa Scognamiglio, and Joacim Tåg. 2025. "JAQ of all trades: Job mismatch, firm productivity and managerial quality." Journal of Financial Economics, 164: 103992.

Dai, Weijia (, Ginger Zhe Jin, and Ben Zou. 2025. "Reducing Nutritional Disparities Through Targeted Pricing: Evidence from a Large Restaurant Chain."

Dencker, John C., and Chichun Fang. 2016. "Rent Seeking and the Transformation of Employment Relationships: The Effect of Corporate Restructuring on Wage Patterns, Determinants, and Inequality." American Sociological Review, 81(3): 467–487.

DiBenigno, Julia. 2022. "How Idealized Professional Identities Can Persist through Client Interactions." Administrative Science Quarterly, 67(3): 865–912.

Doeringer, Peter B., and Michael J. Piore. 1971. Internal labor markets and manpower analysis. Lexington, Mass:Heath.

Drenik, Andres, Simon Jäger, Pascuel Plotkin, and Benjamin Schoefer. 2023. "Paying Outsourced Labor: Direct Evidence from Linked Temp Agency-Worker-Client Data." The Review of Economics and Statistics, 105(1): 206–216.

Faber, Benjamin, and Thibault Fally. 2022. "Firm Heterogeneity in Consumption Baskets: Evidence from Home and Store Scanner Data." The Review of Economic Studies, 89(3): 1420–1459.

Favaron, Saverio Dave, Giada Di Stefano, and Rodolphe Durand. 2022. "Michelin Is Coming to Town: Organizational Responses to Status Shocks." Management Science, 68(9): 6925–6949.

Feigenbaum, James, and Daniel P. Gross. 2024. "Organizational and Economic Obstacles to Automation: A Cautionary Tale from AT&T in the Twentieth Century." Management Science, 70(12): 8520–8540.

Feldberg, Alexandra C. 2022. "The Task Bind: Explaining Gender Differences in Managerial Tasks and Performance." Administrative Science Quarterly, 67(4): 1049–1092.

Frías, Judith A., David S. Kaplan, Eric Verhoogen, and David Alfaro-Serrano. 2024. "Exports and Wage Premiums: Evidence from Mexican Employer-Employee Data." The Review of Economics and Statistics, 106(2): 305–321.

Gartenberg, Claudine, and Julie Wulf. 2020. "Competition and Pay Inequality Within and Between Firms." Management Science, 66(12): 5925–5943.

Gong, Jie, and I. P. L. Png. 2023. "Automation Enables Specialization: Field Evidence." Management Science.

Gottlieb, Joshua D., David Hémous, Jeffrey Hicks, and Morten G. Olsen. 2023. "The Spillover Effects of Top Income Inequality."

Hall, Robert E., and Alan B. Krueger. 2012. "Evidence on the Incidence of Wage Posting, Wage Bargaining, and On-the-Job Search." American Economic Journal: Macroeconomics, 4(4): 56–67.

Ichniowski, Casey, Kathryn Shaw, and Giovanna Prennushi. 1997. "The Effects of Human Resource Management Practices on Productivity: A Study of Steel Finishing Lines." The American Economic Review, 87(3): 291–313.

Jaravel, Xavier. 2019. "The Unequal Gains from Product Innovations: Evidence from the U.S. Retail Sector." The Quarterly Journal of Economics, 134(2): 715–783.

Kelley, Maryellen R. 1990. "New Process Technology, Job Design, and Work Organization: A Contingency Model." American Sociological Review, 55(2): 191–208.

Kohlhepp, Jacob. 2024. "The Inner Beauty of Firms."

Kremer, Michael. 1993. "The O-Ring Theory of Economic Development." The Quarterly Journal of Economics, 108(3): 551–575.

Krueger, Alan B., and Lawrence H. Summers. 1988. "Efficiency Wages and the Inter-Industry Wage Structure." Econometrica, 56(2): 259–293.

Levinthal, Daniel A. 1997. "Adaptation on Rugged Landscapes." Management Science, 43(7): 934–950.

Lounsbury, Michael. 2001. "Institutional Sources of Practice Variation: Staffing College and University Recycling Programs." Administrative Science Quarterly, 46(1): 29–56.

MacDuffie, John Paul. 1995. "Human Resource Bundles and Manufacturing Performance: Organizational Logic and Flexible Production Systems in the World Auto Industry." ILR Review, 48(2): 197–221.

Martin-Caughey, Ananda. 2021. "What's in an Occupation? Investigating Within-Occupation Variation and Gender Segregation Using Job Titles and Task Descriptions." American Sociological Review, 86(5): 960–999.

Massenkoff, Maxim, and Nathan Wilmers. 2023. "Rubbing Shoulders: Class Segregation in Daily Activities."

Ma, Xiao, Marc-Andreas Muendler, and Alejandro Nakab. 2025. "Exporting, Wage Profiles, and Human Capital: Evidence from Brazil." The Review of Economics and Statistics, 107(5): 1371–1387.

Miner, Anne S. 1987. "Idiosyncratic Jobs in Formalized Organizations." Administrative Science Quarterly, 32(3): 327–351.

Narayanan, Sriram, Sridhar Balasubramanian, and Jayashankar M. Swaminathan. 2009. "A Matter of Balance: Specialization, Task Variety, and Individual Learning in a Software Maintenance Environment." Management Science, 55(11): 1861–1876.

Oliva, Rogelio, and John D. Sterman. 2001. "Cutting Corners and Working Overtime: Quality Erosion in the Service Industry." Management Science, 47(7): 894–914.

Osterman, Paul. 2018. "In Search of the High Road: Meaning and Evidence." ILR Review, 71(1): 3–34.

Osterman, Paul, ed. 2020. Creating Good Jobs: An Industry-Based Strategy. The MIT Press.

Podolny, Joel M. 1993. "A Status-Based Model of Market Competition." American Journal of Sociology, 98(4): 829–872.

Rahmandad, Hazhir. 2019. "Interdependence, Complementarity, and Ruggedness of Performance Landscapes." Strategy Science, 4(3): 234–249.

Rahmandad, Hazhir, and Zeynep Ton. 2020. "If Higher Pay Is Profitable, Why Is It So Rare? Modeling Competing Strategies in Mass Market Services." Organization Science, 31(5): 1053–1071.

Shapiro, Carl, and Joseph E. Stiglitz. 1984. "Equilibrium Unemployment as a Worker Discipline Device." The American Economic Review, 74(3): 433–444.

Sherman, Rachel. 2011. "Beyond interaction: customer influence on housekeeping and room service work in hotels." Work, Employment and Society, 25(1): 19–33.

Song, Jae, David J Price, Fatih Guvenen, Nicholas Bloom, and Till von Wachter. 2019. "Firming Up Inequality." The Quarterly Journal of Economics, 134(1): 1–50.

Staats, Bradley R., and Francesca Gino. 2012. "Specialization and Variety in Repetitive Tasks: Evidence from a Japanese Bank." Management Science, 58(6): 1141–1159.

Talbert, Joan, and Christine E. Bose. 1977. "Wage-Attainment Processes: The Retail Clerk Case." American Journal of Sociology, 83(2): 403–424.

Tan, Vaughn. 2015. "Using Negotiated Joining to Construct and Fill Open-ended Roles in Elite Culinary Groups." Administrative Science Quarterly, 60(1): 103–132.

Ton, Zeynep. 2014. The good jobs strategy: how the smartest companies invest in employees to lower costs and boost profits. Boston:New Harvest/Houghton Mifflin Harcourt.

Trost, Jan E. 1986. "Statistically nonrepresentative stratified sampling: A sampling technique for qualitative studies." Qualitative Sociology, 9(1): 54–57.

Wasi, Nada, and Aaron Flaaen. 2015. "Record Linkage Using Stata: Preprocessing, Linking, and Reviewing Utilities." The Stata Journal, 15(3): 672–697.

White, Harrison C. 1981. "Where Do Markets Come From?" American Journal of Sociology, 87(3): 517–547.

Wilmers, Nathan. 2017. "Does Consumer Demand Reproduce Inequality? High-Income Consumers, Vertical Differentiation, and the Wage Structure." American Journal of Sociology, 123(1): 178–231.

Wilmers, Nathan. 2020. "Job Turf or Variety: Task Structure as a Source of Organizational Inequality." Administrative Science Quarterly, 65(4): 1018–1057.

Wilmers, Nathan, and Clem Aeppli. 2021. "Consolidated Advantage: New Organizational Dynamics of Wage Inequality." American Sociological Review, 86(6): 1100–1130.

Wrzesniewski, Amy, and Jane E. Dutton. 2001. "Crafting a Job: Revisioning Employees as Active Crafters of Their Work." Academy of Management Review, 26(2): 179–201.

A QCEW Blanket-Access States

Although the QCEW collects data from all states, some states do not provide unrestricted access to researchers. As a result, these states have been excluded from my analysis. Table A.1 details the states and territories included in this study.

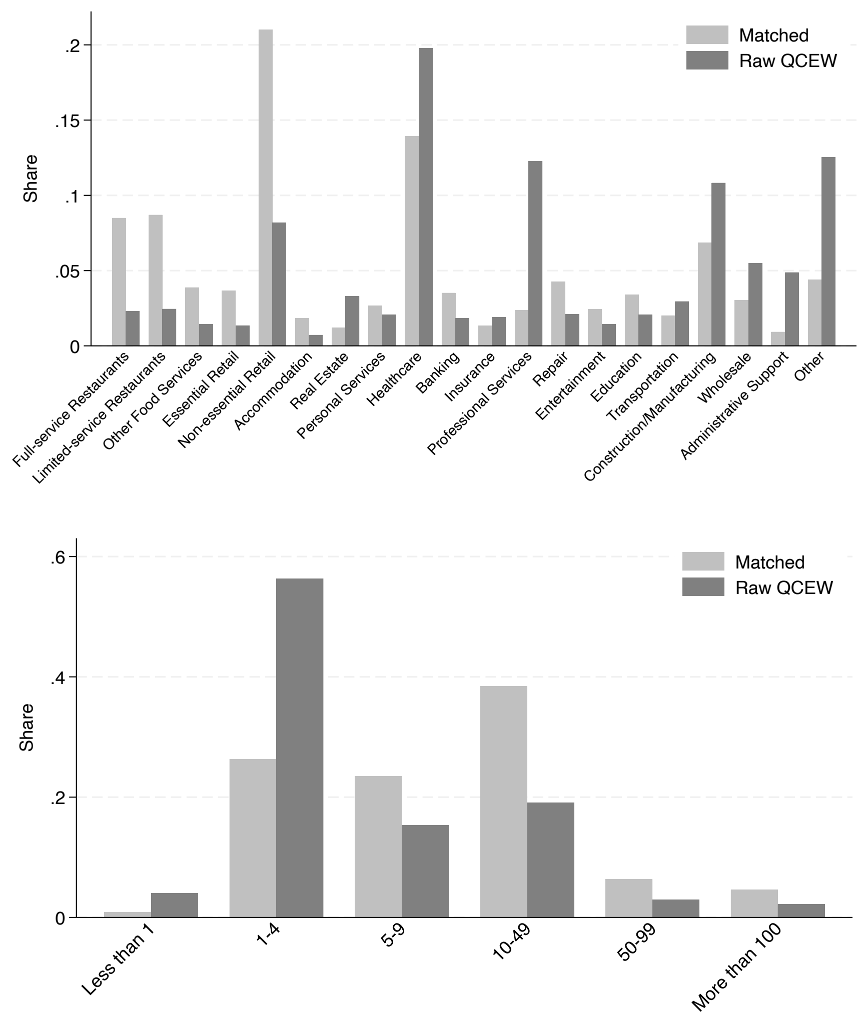
Table A.1: List of Included States and Territories

Alaska	Iowa	New Jersey	Texas
Alabama	Idaho	New Mexico	Utah
Arizona	Indiana	Nevada	Virginia
California	Kansas	Ohio	Virgin Islands
Connecticut	Maryland	Oklahoma	Vermont
Washington D.C.	Maine	South Carolina	Washington
Delaware	Minnesota	South Dakota	Wisconsin
Georgia	Missouri	Tennessee	West Virginia
Hawaii	Montana		

B Additional Details on SafeGraph-QCEW Match

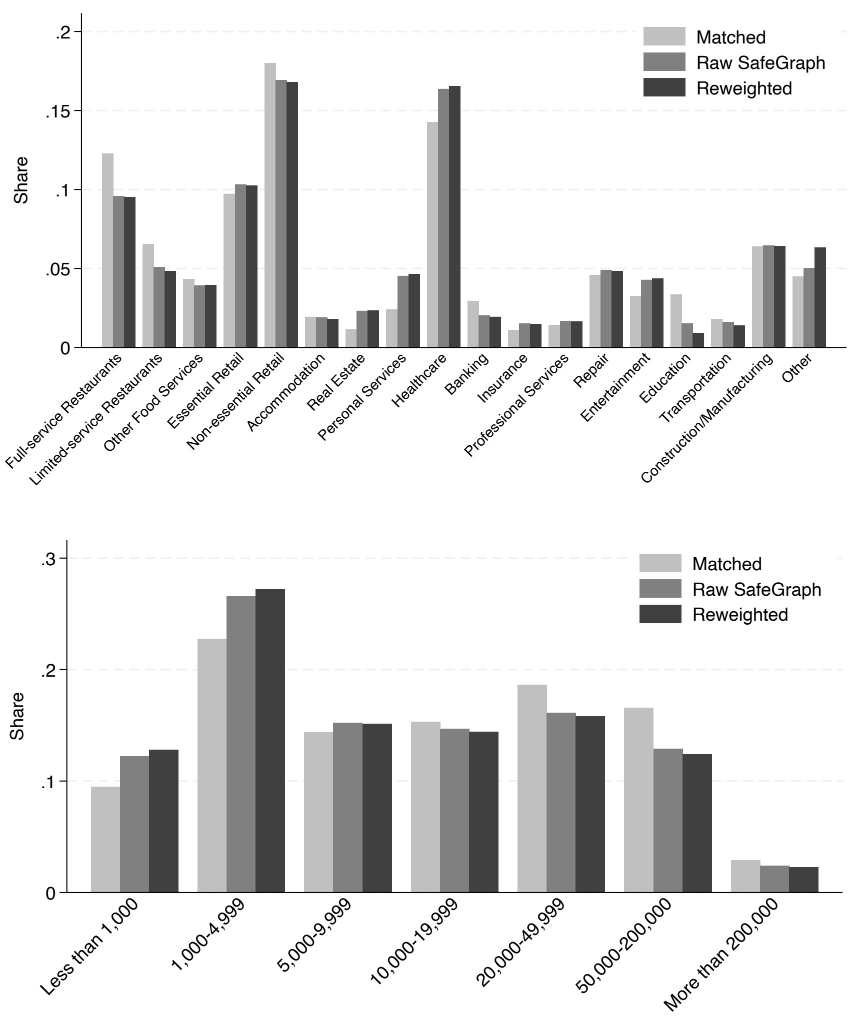
SafeGraph's Places dataset provides POI names and addresses, while QCEW collects establishment names and address information for U.S. businesses with more than three employees through the Annual Refiling Survey (ARS). To match these two datasets by establishment names and addresses, I used the user-written Stata package *reclink2* (Wasi and Flaaen 2015). Initially, I performed exact matches based on standardized workplace names and addresses, using pattern files included with the *reclink2* package (Wasi and Flaaen 2015). Following this, I conducted a series of fuzzy matches with the *reclink2* command, iteratively refining the process to minimize false positives and false negatives.

Figure A.1: Unmatched sample share by industry and number of quarterly visitors



The overall match rate is approximately 36% from the SafeGraph side and 20% from the QCEW side. The lower match rate from the QCEW side is expected, as many of the SafeGraph POIs are customer-facing businesses. However, even the SafeGraph match rate is not as high as desired, primarily due to missing addresses in QCEW. QCEW address information, sourced from ARS, excludes smaller businesses and only surveys one-third of its sampling frame annually. This aligns with the pattern in Figure A.1, which shows that smaller businesses, particularly those with fewer than five employees, are underrepresented in the matched dataset.

Figure A.2: Unmatched sample share by industry and number of quarterly visitors



On the other hand, as shown in Figure A.2, the disparity between matched and raw samples is less pronounced from the SafeGraph side.[7] This is somewhat reassuring given the low match rate, though there are still subtle differences between the matched and raw SafeGraph samples. Matched workplaces tend to attract more customers, with restaurants being overrepresented, while real estate, personal services, and healthcare are also underrepresented. To check whether these differences in size and industry distribution affect the main findings of the paper, I conduct a reweighting exercise described in Appendix C. Figure A.2 also includes the weighted distribution of the matched sample alongside the distributions of the matched and raw samples.

C Robustness Checks on the SafeGraph-QCEW Analysis

Table A.2: Robustness Checks on the Main Cross-sectional Pay Regression

	(1)	(2)	(3)	(4)	(5)
Visitor Income	0.064***	0.076***	0.054***	0.055***	0.074***
	(0.004)	(0.007)	(0.005)	(0.012)	(0.005)
Fixed effects:					
Time X NAICS X CBG	×	×	×	×	×
Controls	×	×	×	×	×
N	3,395,868	1,153,982	3,395,868	370,205	2,932,432

* Dependent variables for all models are logged average pay at establishment level from QCEW. Pay deflated with 2017 dollars. Visitor income is median visitor income predicted from SafeGraph with 2016-2020 combined ACS. Time is at year-by-quarter-level. Controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees. Column 1 is the main model from Table 2 Column (4). Column 2 only uses workplaces where more of their visitors are from neighborhoods with homogeneous income levels. Column 3 shows the weighted regression results that corrects for the changes in industry and visitor size shares from data merge. Column 4 shows the results from the analysis only using 2022 sample. Column 5 drops healthcare and manufacturing sectors. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

[7]Note that the industry distribution of the matched sample differs from Figure 2, as Figure A.2 represents unique places, while Figure 2 is at the place-by-time level.

Table A.2 presents the results from additional robustness checks for the main cross-sectional regression analysis in Column (1), which is equivalent to the results in Column (4) of Table 2. In Column (2), I use only those workplaces whose visitors are more likely from CBGs with homogeneous income levels. Since I approximate median visitor income across CBG-level averages from ACS, the value will become more accurate as the income levels within each CBG converge. I calculate establishment-level averages of within-CBG income variance across those CBGs where the visitors reside. Then, I use only those establishments with average within-CBG income variance lower than mean to replicate the model in Column (1). The coefficient increases as expected, suggesting possible measurement errors in visitor income driving attenuation bias.

Column (3) illustrates the results from a weighted regression, where I have reweighted each workplace by the ratio of the number of raw SafeGraph observations to matched sample observations in each industry by visitor number cell that the workplace is in. These cell definitions are consistent to what has been defined in Figure A.2. I try to address possible selection bias (see Appendix B for more discussion) coming from the merge process this way. Comparing the result with the main model in Column (1), while the coefficient decreases a little, a 10% increase in visitor income still predicts a statistically significant 0.5% increase in average worker pay.

Column (4) only uses observations from 2022 to replicate the analysis in Column (1). While the SafeGraph dataset dates as far back as 2018, it is less reliable for the earlier periods as the number of mobile devices was smaller. On the other hand, visit patterns could have been contaminated by irregularities during the COVID-19 pandemic when there were frequent lockdowns or store closures. The 2022 data is free from both of these concerns. The result is again consistent with the main model, albeit the coefficient being a bit smaller.

Lastly, Column (5) drops the construction, manufacturing, and healthcare industries, which could be problematic for our analysis. While SafeGraph *visitors* have been assumed to represent *consumers* in the analyses, this assumption likely holds primarily for service industries. Additionally, wage rates in the healthcare sector are largely set by government and insurance companies (Osterman 2020) rather than at the establishment level. Including these industries may therefore bias the true correlation between consumer income and pay levels. The result in Column (5) demonstrates a higher coefficient when these industries are excluded from the sample.

D Industry and Occupation Classifications

Figure 4 replicates the main cross-sectional pay regressions in Table 2, only using establishments from specific industries. The definition of industries is as below, following Massenkoff and Wilmers 2023.

- Full-service Restaurants: NAICS Code 722511
- Limited-service Restaurants: NAICS Code 722513
- Other Food Services: NAICS Codes 722000-722999
- Essential Retail: NAICS Codes 445000-447999
- Non-essential Retail: NAICS Codes 440000-459999
- Accommodation: NAICS Codes 721000-721999
- Real Estate: NAICS Codes 531000-531999
- Personal Services: NAICS Codes 812000-812999
- Healthcare: NAICS Codes 620000-629999
- Banking: NAICS Codes 522000-522999
- Insurance: NAICS Codes 524000-524999

- Professional Services: NAICS Codes 540000-549999
- Repair: NAICS Codes 811000-811999
- Entertainment: NAICS Codes 710000-719999
- Education: NAICS Codes 610000-619999
- Transportation: NAICS Codes 480000-499999
- Construction: NAICS Codes 230000-239999
- Manufacturing: NAICS Codes 310000-339999
- Wholesale: NAICS Codes 420000-429999
- Administrative Support: NAICS Codes 561000-561999

- Managerial: 2-digit SOC Code 11 (Management)
- Professional: 2-digit SOC Codes 13 (Business and Financial), 15 (Computer and Math), 17 (Engineering), 19 (Science), 23 (Legal)
- Frontline: 2-digit SOC Codes 31 (Healthcare Support), 35 (Food Services), 39 (Personal Services), 41 (Sales), 43 (Administrative Support), 47 (Construction and Extraction), 51 (Production), 53 (Transportation and Material Moving)

E OEWS match

The Occupational Employment and Wage Statistics (OEWS) dataset, a restricted-access resource from the Bureau of Labor Statistics, provides workplace-by-occupation employment and wage statistics collected through representative, repeated surveys. OEWS uses a three-year sampling strategy, where the establishments surveyed over a three-year period form a representative sample. Under this design, an establishment sampled in year y is not surveyed again until $y + 3$. Each year, OEWS surveys about 400,000 establishments, resulting in a combined three-year sample of roughly 1,200,000 establishments.

I make two additional adjustments to the OEWS dataset. First, I create 5-digit occupation codes to harmonize the various 6-digit occupational classification systems used during the transition to the new Standard Occupational Classification (SOC) system in 2018. Second, I apply the methodology from Wilmers and Aeppli (2021) to convert the wage variable to a continuous scale, as most of the OEWS wage data was reported in intervals prior to 2020.

I merged the OEWS data with the SafeGraph-QCEW dataset using workplace-level identifiers common to both OEWS and QCEW. The merge process drops a large number of QCEW establishments as OEWS surveys roughly 180,000 establishments annually, while QCEW covers the full universe of establishments across more than half the states. On the other hand, the state restriction of the QCEW sample leaves a significant portion of OEWS establishments unmatched (see Appendix A). I further drop manufacturing and healthcare industries from the matched sample used for analysis: the sampling strategy of OEWS is biased towards larger establishments with more workers, leading to the overrepresentation of these two problematic industries (see Appendix C for more discussion on why these could be problematic).

The final SafeGraph-QCEW-OEWS dataset is structured at the Quarter-by-Workplace-by-Occupation level, with a total of 2,130,753 observations and 286,565 Quarter-by-Establishment pairs. Table A.3 illustrates the descriptive statistics of this sample.

F Additional Results from Occupational Analyses

First, I examine whether variations in occupational composition drive pay differences across establishments. To test this possibility, I estimate the relationship between establishment-level average pay and visitor income, both with and without controlling for occupational composition. To

Table A.3: Descriptive Statistics of OEWS matched sample (N=2,130,753)

Variable	Mean	S.D.	Min.	Max.
log(occupational wage)	2.61	0.55	1.57	4.82
log(Median visitor income)	10.99	0.28	8.29	12.42
log(Average employment)	3.97	1.43	-1.09	10.67
log(Number of visitors)	8.25	1.85	1.81	14.29
Share Low wage Occ.	0.41		0	1
Share Mid. wage Occ.	0.30		0	1
Share High wage Occ.	0.28		0	1
Share Managerial Occ.	0.04		0	1
Share Frontline Occ.	0.69		0	1
Share Supervisor Occ.	0.06		0	1
Share Customer-facing Occ.	0.43		0	1

Note: Observations at Time X Workplace X Occupation level, 2018-2022.

construct a measure of workplace-level occupational composition, I categorize occupations into ten groups based on their 2012 OEWS pay levels, accounting for their distribution across different workplaces. I assign pay group numbers to each five-digit SOC occupation, with 10 representing the highest-paid occupations and 1 the lowest. Empirically, I derive these decile groups by estimating occupation fixed effects from a two-way fixed effects model of wages on five-digit occupations and workplace identifiers, following Wilmers and Aeppli (2021). Using this approach, I compute the average occupational pay group number for each SafeGraph-QCEW-OEWS matched workplace, which serves as its occupational composition score.[8]

I estimate a series of workplace-level regressions using the SafeGraph-QCEW-OEWS matched sample, varying the granularity of regional fixed effects to account for the smaller sample size resulting from the OEWS coverage restriction.[9] If pay differences across establishments are primarily

[8]I use pay deciles for occupations instead of raw occupation fixed effects, as decile-based scores better capture the boundaries between occupations. This distinction is particularly important given that wages in the OEWS data are reported in intervals, which may cause fixed effects to imperfectly or biasedly reflect underlying wage differences.

[9]OEWS is based on a three-year rotating survey design, sampling approximately 400,000 establishments per year. More details are provided in Appendix E.

driven by differential composition of occupations, then controlling for the score should systematically attenuate the effect of visitor income on pay levels across different model specifications.

Table A.4: Cross-sectional Pay Regressions, Controlling for Occupational Composition

	(1)	(2)	(3)	(4)	(5)	(6)
Visitor Income	0.046***	0.040***	0.079	0.101*	0.130	0.126
	(0.012)	(0.012)	(0.055)	(0.054)	(0.115)	(0.115)
Occ. Composition Score		0.095***		0.068***		0.060***
		(0.001)		(0.006)		(0.011)
Fixed effects:						
Time X ZIP Code	×	×				
Time X NAICS	×	×				
Time X NAICS X ZIP Code			×	×		
Time X NAICS X CBG					×	×
Controls	×	×	×	×	×	×
N	222,778	222,778	14,213	14,213	4,007	4,007

* Observations at 2018-2022 workplace level, matching SafeGraph-QCEW-OEWS and dropping healthcare and manufacturing industries. Pay deflated with 2000 dollars. Visitor income is median visitor income predicted from SafeGraph with 2016-2020 combined ACS. Time is at year-by-quarter-level. Controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees. The Occupation Composition Score is the workplace-level average of globally calculated occupation pay deciles, which group occupations into 10 categories based on their 2012 average wages. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

The results provide limited evidence that occupational composition accounts for the relationship between visitor income and workplace pay. The modest decline in point estimates after controlling for occupation scores indicates that occupational differences explain only about 3% of the visitor income effect on pay in models with CBG fixed effects. Across various specifications comparing different samples, controlling for occupational composition score captures only a small portion, if any, of the visitor income effect on pay.

Motivated by this finding, I next estimate occupation-level regressions with and without occupation fixed effects. Including occupation fixed effects enables comparisons among workers in the same occupation but employed at establishments serving different consumer bases within the

Table A.5: Cross-sectional Pay Regressions at Occupation level

	(1)	(2)	(3)	(4)
Visitor Income	0.046	0.041	0.053	0.070
	(0.033)	(0.025)	(0.060)	(0.044)
Fixed effects:				
Time X NAICS X CBG	×		×	
Time X NAICS X CBG X Occ.		×		×
Controls			×	×
N	527,643	23,216	506,950	22,333

* Observations at 2018-2022 Time X Workplace X 5-digit Occupation level, matching SafeGraph-QCEW-OEWS and dropping healthcare and manufacturing industries. Dependent variables for all models are logged occupational wage predicted from OEWS via midpoint Pareto method following Wilmers and Aeppli 2021. Pay deflated with 2000 dollars. Visitor income is median visitor income predicted from SafeGraph with 2016-2020 combined ACS. Time is at year-by-quarter-level. Controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees.
* Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

same neighborhood and industry. If higher consumer income predicts higher pay primarily through increased payoffs to each occupation – rather than through shifts in occupational composition – one would not expect a substantial reduction in the visitor income coefficient after including occupation fixed effects. The results in Table A.5 show little reduction or a slight increase in the coefficient for visitor income, indicating that the positive association between consumer income and worker pay is largely driven by within-occupation pay differences rather than by changes in occupational composition across establishments.

G Glassdoor match

Glassdoor is an online platform where users voluntarily share pay information for their current or past jobs, along with short text reviews. For this study, I utilize only the pay-related data from Glassdoor and match it with the SafeGraph visitor income variable. The Glassdoor dataset includes various compensation components, such as base salary, cash and stock bonuses, profit sharing, sales commissions, and tips. I consolidate these components into a single logged hourly pay variable. Because each compensation report is associated with the year in which the payment was received, annual aggregation is the most natural approach. For the event-study analyses in Section 3.3.3, however, more granular timing is preferable. I therefore restrict the sample to reviews submitted in the same year as the reported compensation and assign each pay observation to the quarter in which the review was posted.

The Glassdoor pay dataset spans from 2006 to 2023, containing over 21 million pay records submitted by individual users, with approximately 40% of observations coming from 2021 onward. However, unlike the SafeGraph dataset, Glassdoor provides regional information only at the city level, and employer data is recorded at the firm level rather than the establishment level. To align the datasets, I aggregate visitor information from SafeGraph to the firm-by-city level and match it with employer information from individual pay reviews in Glassdoor. Specifically, I use city, industry, and employer name variables to conduct a fuzzy match using the Stata module *reclink2* (Wasi and Flaaen 2015).

For analyses that include worker fixed effects, such as the examination of worker selection in Section 4.3.3, it is important to use the longer time window available in the Glassdoor data to observe a sufficient number of worker moves. I therefore construct a time-invariant indicator of whether a firm serves high-income consumers at the firm-by-city level. The construction of this variable follows the procedure used for the job-postings data in Section 4.1. First, I compute the average consumer income quantile for each firm-by-city in the SafeGraph dataset. Specifically, I first calculate each unit(firm-by-city)'s pay decile while controlling for time, industry, and city, then average these quantiles within each unit over time. I then define a binary indicator for a high-income consumer base, assigning a value of 1 if the average decile value exceeds 5.5. This definition assumes that most units maintain a stable consumer base over time, with minimal shifts between high- and low-income segments. At the same time, because the analysis is conducted at the city level, the uneven worker sorting that was a concern in the job-postings data is less problematic: more than 75% of establishments remain distinct rather than being collapsed with other establishments at the firm-by-city level. For the difference-in-differences analyses in Section 3.3.3, I instead use a time-varying measure of average visitor income aggregated across establishments within each firm-by-city, weighting establishments by their 2019 visitor counts.

After the initial matching between Glassdoor and SafeGraph, I retain only those firm-by-city units that exceed a certain match score threshold. This threshold is determined by ensuring that at least 95% of sampled matches are correct, based on a random sample of 100 observations above the threshold.

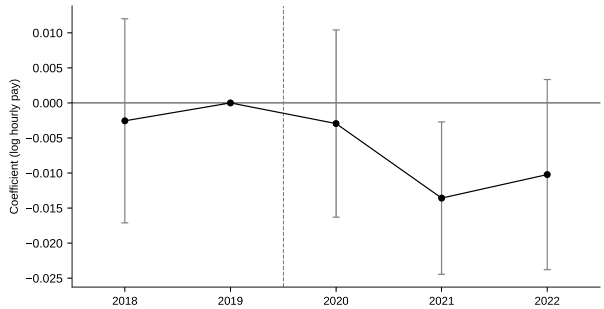
The final dataset comprises 2,241,750 pay reviews from 232,410 firm-by-city units spanning the years 2006–2023. Table A.6 presents the descriptive statistics for the key variables used in the analyses.

Table A.6: Descriptive Statistics of Matched SG-Glassdoor Sample (N=2,241,750)

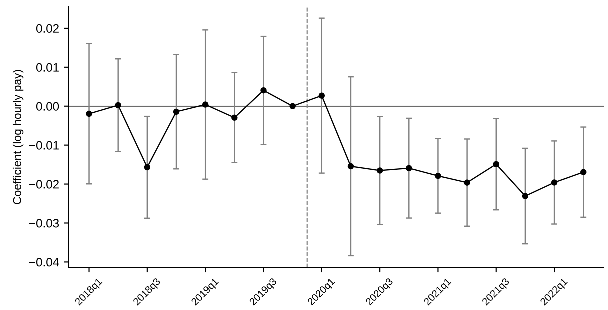
Variable	Mean	S.D.	Min.	Max.
log(Hourly pay)	2.99	0.64	1.00	5.99
log(Number of visitors)	8.43	1.91	1.81	14.12
log(Number of employees)	9.32	2.56	0	15.42
Years of relevant experience	4.42	5.64	0	60
High-income consumers	0.58	0.49	0	1

Note: Each observation represents a pay review submitted by users in Glassdoor and is linked to SafeGraph visitor information at establishment level. Pay includes both base and extra pay. The dummy variable for high-income consumer base and the number of visitors from SafeGraph averaged across time within each establishment. Number of employees recorded at firm level from Glassdoor.

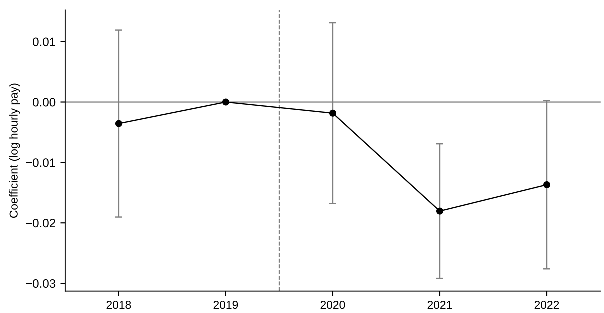
H Additional Results for the Commuter-Shock Design



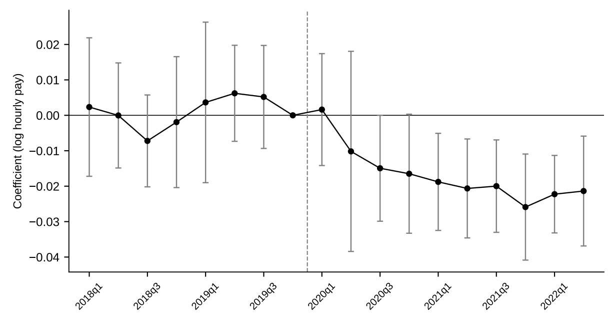
(a) Occ.×Qtr FE, within city, yearly



(b) Occ.×Qtr FE, within metro area, quarterly



(c) No traffic strata, within city, yearly



(d) No traffic strata, within metro area, quarterly

Figure A.3: Robustness of the pay response

*All panels re-estimate the pay event studies of Figure 6 on the same sample (662,408 pay reports at 98,903 matched establishments), with the same controls and clustering; left panels use the yearly within-city specification (base 2019), right panels the quarterly within-metropolitan-area specification (base 2019q4). Panels (a) and (b) add occupation×quarter fixed effects. Panels (c) and (d) drop the traffic-change-quintile×quarter fixed effects while retaining the contemporaneous log-visitor-count control. Bars are 95% confidence intervals; the dashed line marks the onset of the pandemic.

I Additional Details on Interviews

I.1 Sample

Table A.7: List of Interviewees

Name	Title	Restaurant	Market Segment
A	Assistant Manager	Willow Table	Casual
B C	Assistant GM Owner	Ember & Sage	Casual Fine Dining
D	Owner	Everton Grill Hearth & Co.	Casual Casual
E	Director of Operations	Stonewood Bellwether Linden Café	Casual Fine Dining Fine Dining Café
F	General Manager	Foxhall	Casual Fine Dining

Note: The names of the interviewees and the names of the restaurants are pseudonyms.

Table A.7 presents the sample of interviews conducted for this study. Potential interviewees were contacted through walk-ins and cold emails. In selecting establishments, I aimed to capture variation in price ranges and target customer segments (Trost 1986). Four of the five interviews were recorded and transcribed, with the exception of Interview D. The interviews took place between December 2024 and January 2025, each lasting approximately 30 minutes and conducted in person.

I.2 Interview Protocols

- How are roles structured within your establishment?
  - Are there certain roles that new employees typically do not take on?
  - How are tasks distributed within each role?
- What distinguishes job applicants who are hired from those who are not?
- How does your pay level compare to your competitors?

  - What factors drive these differences?
  - Do these differences vary by role?
- How do employees transition between roles?
  - Does this happen within a single workday as well?
  - What is the reasoning behind these role transitions?
  - How do employees progress to supervisory positions?
- How significant is the difference between highly skilled workers and those with less experience?
  - Does this difference primarily stem from the hiring process or training?
  - What is the average tenure of employees in your establishment?
  - Do you offer internal training programs? How important are they to your business?
- What proportion of your workforce is part-time?
  - How do their job responsibilities compare to those of full-time employees?
  - How does their compensation differ from full-time employees?
  - Are there specific challenges associated with employing part-time workers compared to full-time staff?
- What aspects of the job make it attractive to employees?
- What are the biggest HR challenges you face?
- Are there differences in expectations for your workers compared to other restaurants?
  - What factors drive these differences?
- Do you have a specific target consumer base?
  - What do you think your customers value the most?
  - How do you tailor your operations to meet these demands?

J Additional Details on SafeGraph-Job postings Match

I match in SafeGraph visitor measures to the job postings dataset using employer names and MSA codes through *reclink2* module from Stata (Wasi and Flaaen 2015). I retain only those units that exceed a certain match score threshold in employer names. This threshold is determined by ensuring that at least 95% of sampled matches are correct, based on a manual review of 100 randomly sampled matches above the cutoff. The final matched sample includes 1,562,393 postings from 24,564 unique firm-by-metro-region units – representing roughly 5% of all SafeGraph establishments aggregated at the MSA level. From the job postings side, among firm-by-metro-region units in the Retail (NAICS 44 and 45) and the Accommodation and Food Services sector (NAICS 72) for which employer information is available, 30% are successfully matched to SafeGraph establishments. Note, however, that industry codes are available for fewer than 10% of firms in the Revelio Labs database, likely biasing the sample used in this match rate calculation toward larger, more publicly visible firms.

The modest match rate reflects three main limitations. First, not all firms post their jobs on Indeed. Second, because Revelio Labs relies on LinkedIn [10] pages for employer information – and LinkedIn primarily covers higher-skill segments of the labor market – the matched sample is skewed toward larger chains. In fact, employer names are missing for about half of the postings in the dataset. Third, unlike the QCEW, which reports both legal and trade names for each establishment, the Revelio Labs dataset includes only a company name, which may not always reflect the business name displayed at the establishment level (*e.g.*, on storefront signage).

[10] LinkedIn is an online platform for professional networking, where individuals create profiles to showcase their work experience, education, and skills – functioning as a dynamic, public-facing resumé. The platform also serves businesses, allowing companies to maintain organizational pages, post job openings, and recruit talent directly through LinkedIn's hiring tools.

K Robustness Checks on the SafeGraph-Job postings Analyses

In the main SafeGraph–job postings matched sample, I collapse multiple establishments belonging to the same firm at the metropolitan or micropolitan statistical area (MSA) level. While this approach facilitates matching, it may introduce bias if establishments within the same firm differ substantially in their consumer bases or employment practices. To address this concern, I present results using a restricted sample that includes only firm × MSA units associated with a single establishment in the SafeGraph data – thus avoiding any aggregation. The findings in Table A.8 and Table A.9 are largely consistent with those in Table 6 and Table 8, respectively, providing support for the robustness of the main results.

Table A.8: Average Task Price Across Jobs, Single-establishment Sample

Granularity of work activities	k = 1000		k = 1500	
	(1)	(2)	(3)	(4)
High-income Consumer Base	0.053***	0.107***	0.026***	0.077***
	(0.005)	(0.005)	(0.006)	(0.005)
ln(Number of Visitors)		×		×
Fixed effects:				
Time X MSA X Ind.	×		×	
Time X MSA X Ind. X Occ.		×		×
N	260,785	236,682	260,785	236,682

* Observations represent individual job postings from Revelio Labs 2022, excluding postings from the manufacturing and healthcare industries. The dependent variable is the standardized average task price associated with each job, constructed using either k = 1000 or k = 1500 task classifications. This variable captures how jobs are composed of tasks with differing market values. Task prices are estimated by regressing logged salaries on task indicators using a 5% random sample of postings. The dummy variable for high-income consumer base is identified with average visitor income, controlling for Time X Region X NAICS fixed effects with 2022 SafeGraph dataset. Regions at each Metropolitan statistical area. Industry and occupation are measured at the 6-digit NAICS and SOC levels, respectively. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

I also use a high-income consumer indicator as the main measure in the job-posting analyses. Establishment-level visitor income is first converted into deciles before being aggregated to the

Table A.9: Number of Distinct Tasks by Consumer Income, Single-establishment Sample

Granularity of tasks	k = 100		k = 1000		k = 1500	
Sample Occ.	(1)	(2)	(3)	(4)	(5)	(6)
High-income Consumer Base	-0.181*** (0.004)	-0.147*** (0.004)	-0.198*** (0.005)	-0.159*** (0.005)	-0.197*** (0.005)	-0.159*** (0.005)
ln(Number of Visitors)		×		×		×
Fixed effects:						
Time X MSA X Ind.	×		×		×	
Time X MSA X Ind. X Occ.		×		×		×
N	260,785	236,682	260,785	236,682	260,785	236,682

* Observations represent individual job postings from Revelio Labs for the year 2022, excluding postings from the manufacturing and healthcare industries. The dependent variable is the logged number of tasks identified from job postings. Results are presented separately based on the task definitions used, clustering job descriptions into either 100, 1000, or 1,500 task categories. The dummy variable for high-income consumer base is identified with average visitor income, controlling for Time X Region X NAICS fixed effects with 2022 SafeGraph dataset. Regions at each Metropolitan statistical area. Industry and occupation are measured at the 6-digit NAICS and SOC levels, respectively.
* Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

firm-by-MSA level, limiting the influence of outlier establishments. Because the resulting average-decile measure lacks a straightforward substantive interpretation, I dichotomize it at 5.5 for the main specifications. The results are nevertheless robust to using the continuous average-decile measure, as shown in Table A.10 and Table A.11, which replicate Table 6 and Table 8, respectively. Indeed, the estimated effects are substantially larger when the continuous measure is used, suggesting that replacing the linear regressor with a binary indicator yields a more conservative specification.

The distributional patterns shown in Figure A.4 provide further support for these results. As with the pay outcomes in Figure 3, consumer income exhibits an approximately linear relationship with both average task price and the number of tasks per posting. The latter relationship is somewhat less linear, however, as it becomes pronounced primarily above the median of the consumer-income distribution.

Table A.10: Average Task Price Across Jobs: Continuous Consumer Income

Granularity of work activities	k = 1000		k = 1500	
	(1)	(2)	(3)	(4)
Avg. Consumer Income Decile	0.0308***	0.0285***	0.0210***	0.0203***
	(0.0005)	(0.0005)	(0.0005)	(0.0005)
ln(Number of Visitors)		×		×
Fixed effects:				
Time X MSA X Ind.	×		×	
Time X MSA X Ind. X Occ.		×		×
N	1,290,976	1,248,986	1,290,976	1,248,986

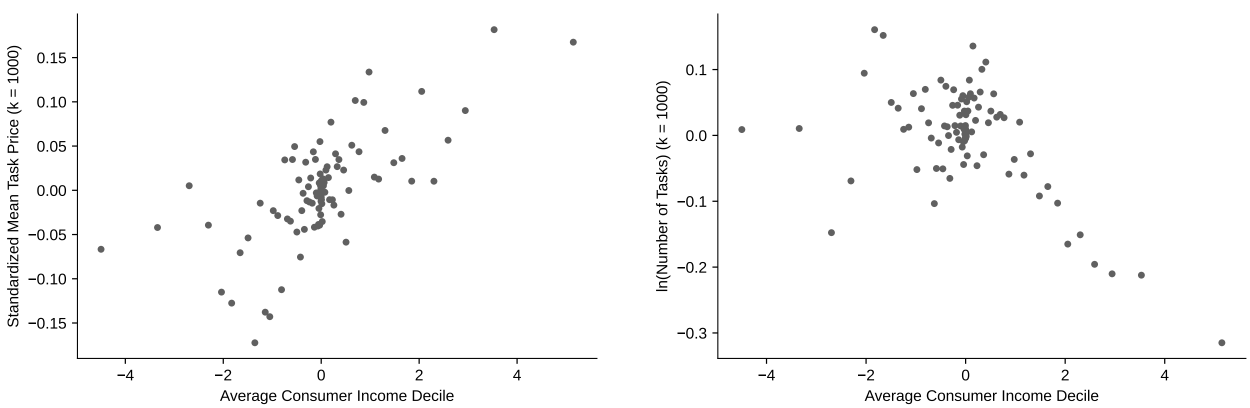
* Observations represent individual job postings from Revelio Labs 2022, excluding postings from the manufacturing and healthcare industries. The dependent variable is the standardized average task price associated with each job, constructed using either k = 1000 or k = 1500 task classifications. This variable captures how jobs are composed of tasks with differing market values. Task prices are estimated by regressing logged salaries on task indicators using a 5% random sample of postings. Average consumer income decile is the continuous measure underlying the high-income dummy in Table 6: the visitor-weighted average income decile (1-10) of each firm-by-metro unit. Regions at each Metropolitan statistical area. Industry and occupation are measured at the 6-digit NAICS and SOC levels, respectively. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

Table A.11: Number of Distinct Tasks by Consumer Income: Continuous Measure

Granularity of tasks	k = 100		k = 1000		k = 1500	
Sample Occ.	(1)	(2)	(3)	(4)	(5)	(6)
Avg. Consumer Income Decile	-0.0292***	-0.0271***	-0.0327***	-0.0287***	-0.0326***	-0.0288***
	(0.0004)	(0.0004)	(0.0005)	(0.0005)	(0.0005)	(0.0005)
ln(Number of Visitors)		×		×		×
Fixed effects:						
Time X MSA X Ind.	×		×		×	
Time X MSA X Ind. X Occ.		×		×		×
N	1,290,976	1,248,986	1,290,976	1,248,986	1,290,976	1,248,986

* Observations represent individual job postings from Revelio Labs for the year 2022, excluding postings from the manufacturing and healthcare industries. The dependent variable is the logged number of tasks identified from job postings. Results are presented separately based on the task definitions used, clustering job descriptions into either 100, 1000, or 1,500 task categories. Average consumer income decile is the continuous measure underlying the high-income dummy in Table 8: the visitor-weighted average income decile (1-10) of each firm-by-metro unit. Regions at each Metropolitan statistical area. Industry and occupation are measured at the 6-digit NAICS and SOC levels, respectively. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

Figure A.4: Binned Scatterplot of Task Price, Number of Tasks, and Visitor Income, Adjusted for Time, Metro, and Industry Effects



* Data from Safegraph matched with Revelio-Indeed job postings, 2022. Controlling for Time × Metro × NAICS. Visitor income predicted with 2016-2020 combined ACS.

L Additional Job Specialization Analysis

Table A.12: Number of Distinct Tasks by Consumer Income, By Industry

Sample Industry	Accomm. & Food		Other Industries	
Granularity of work activities	$k = 100$	$k = 1500$	$k = 100$	$k = 1500$
High-income Consumer Base	-0.066***	-0.097***	0.024***	0.070***
	(0.002)	(0.003)	(0.002)	(0.002)
ln(Number of Visitors)	×	×	×	×
Fixed effects:				
Time X MSA X Ind. X Occ.	×	×	×	×
N	442,824	442,824	806,162	806,162

* Observations represent individual job postings from Revelio Labs for the year 2022, excluding postings from the manufacturing and healthcare industries. The dependent variable is the logged number of tasks identified from job postings. Results are presented separately based on the task definitions used, clustering job descriptions into either 100 or 1,500 task categories. The Accommodation and Food Services sector is defined using the 2-digit NAICS code 72. The dummy variable for high-income consumer base is identified with average visitor income, controlling for Time X Region X NAICS fixed effects with 2022 SafeGraph dataset. Regions at each Metropolitan statistical area. Industry and occupation are measured at the 6-digit NAICS and SOC levels, respectively. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

M SafeGraph-Glassdoor Analysis

In Glassdoor, users indicate whether their compensation is reported on an hourly, monthly, or annual basis and separately disclose pay components such as base pay, bonuses, tips, and profit sharing. This reporting structure allows me to assess whether differences in work hours or pay composition underlie the correlation documented in Section 3.3.1. Specifically, I first examine the relationship between consumer income and worker pay using hourly pay measure derived from Glassdoor. I then assess whether this relationship is driven by base pay or by any non-base pay components. In the latter analysis, I restrict the sample to workers who report extra pay, as these users constitute a smaller subsample and may differ systematically in reporting behavior.

Table A.13: Consumer Income and Worker Pay: Hourly Pay, Base Pay, and Extra Pay

Sample	All	With Extra Pay		
Dependent Variable	Total	Total	Base Only	Extra Only
High-inc. Consumer	0.040***	0.020***	0.026***	-0.013
	(0.001)	(0.003)	(0.003)	(0.011)
Fixed effects:				
Yr. X City X NAICS	×	×	×	×
Controls	×	×	×	×
N	2,032,136	397,427	397,427	397,427

* Observations at 2018-2023 year by worker level, matching SafeGraph visitor income data to Glassdoor pay reviews data. I replicate the analysis using the full sample and the subsample of workers who report extra pay. All dependent variables are measured in hourly terms and logged, using three alternative pay measures. *Total* includes base pay and additional compensation, including salaries, wages, bonuses, and tips. *Base only* includes salaries and wages, excluding bonuses and tips. *Extra only* includes bonuses and tips only. The dummy variable for high-income consumer base is defined at firm level, controlling for Time X City X NAICS cells. Controls include logged visitor counts, logged firm size, and employment status. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

The estimates in Table A.13 indicate that high-income consumers are consistently associated with higher worker pay, even after accounting for potential differences in work hours. Moreover, the results suggest that base pay accounts for most of this relationship. Taken together, these patterns imply that the findings in Section 3.3.1 are unlikely to be driven solely by differences in work hours or by tips.

Next, I examine whether worker selection can account for this relationship by controlling for unobserved, time-invariant worker characteristics. To do so, the sample used for Table A.14 is restricted to individuals who submitted multiple job reviews, allowing the inclusion of individual fixed effects. These observations constitute approximately 15% of the original Glassdoor pay dataset. The results indicate that worker fixed effects absorb most of the pay variation previously attributed to consumer income, suggesting that worker selection plays a central role in explaining pay differences across firms serving consumers with different income levels.

Table A.14: Log hourly pay regressed on consumer base, with and without individual fixed effects

	(1)	(2)	(3)	(4)
High-inc. Consumer	0.022**	0.006	0.014	0.005
	(0.011)	(0.006)	(0.009)	(0.006)
Fixed effects:				
Yr. X City X NAICS	×	×	×	×
Individual		×		×
Yrs. Experience			×	×
Other Controls	×	×	×	×
N	180,406	180,406	180,392	180,392

* Observations at 2018-2023 year by worker level, matching SafeGraph visitor income data to Glassdoor pay reviews data. Dependent variable is logged hourly pay that workers reported for a job, which includes all salary, wages, bonuses, and tips. The dummy variable for high-income consumer base is defined at firm level, controlling for Time X City X NAICS cells. Models (3) and (4) include flexible controls for the years of relevant experience. Other controls include logged visitor counts, logged firm size, and employment status. Standard errors clustered at firm level. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.